Highlights

Multi-Agent Reinforcement Learning for Surface–Subsurface Observation of Tropical Cyclones Using Submersible Unmanned Vessels

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- Dual-mode SUVs track surface winds and sample subsurface variables.
- Historical atmosphere–ocean environments cover 400 TCs in the western North Pacific.
- Parameter-sharing IPPO is trained for TC observation with 4–16 SUVs.
- Comparison between MARL and traditional path-planning algorithms at both stages.

Multi-Agent Reinforcement Learning for Surface–Subsurface Observation of Tropical Cyclones Using Submersible Unmanned Vessels

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ABSTRACT

Tropical cyclones are destructive, while intensity and rapid-intensification forecasts remain limited by sparse near-core observations. Conventional platforms rarely combine rapid transit with surface-wind and subsurface-current sensing. Blue Whale-class submersible unmanned vessels can transit rapidly at the surface to TC cores, then dive to observe temperature, salinity, currents, and biogeochemical variables. We use simulation-based multi-agent reinforcement learning for autonomous observation and navigation. The environments combine ERA5 10 m winds and HYCOM currents from 0 to 60 m with IBTrACS tracks for 400 western North Pacific TCs. A shared-observation independent proximal policy optimization (IPPO) policy outputs one joint formation action for all surface SUVs and maps local current histories to in-vehicle commands underwater at 10 min intervals. Surface SUVs therefore share observations and preserve their initial relative positions exactly; subsurface SUVs use only local current histories. We compare 12 m/s surface and 1.5 m/s submerged limits, close to current Blue Whale design speeds, with 20 m/s and 2.0 m/s as future upgrades. Compared with four traditional planners at surface experiments, the IPPO policy achieves the lowest final tracking error, which means that the SUVs can reach the TC core region more readily and obtain more valuable observations. For the surface IPPO policy, we further evaluate ring, line, grid, and V-shaped formations. After reaching the TC core region, the SUVs dive vertically to different depths from 0 to 60 m for further observations. Underwater, IPPO attains the largest geometric and current-weighted coverage, while the potential field algorithm attains the highest mean experienced current. Limitations remain that training and validation are conducted in a simulation environment built from historical reanalysis data; real-vehicle experiments are not included and require future work.

1. Introduction

Tropical cyclones (TCs) threaten coastal communities, infrastructure, and offshore operations. Climate assessments indicate that some TC changes are detectable despite observational uncertainty [1] and project an increasing proportion of very intense TCs under continued warming [2], while global TC frequency may decline [3] and rapid-intensification events have recently increased in offshore regions [4]. Landfalling TCs over the western North Pacific and China have also become more destructive, with increasing nearshore intensification and exposure [5, 6, 7, 8]. Track forecasts have improved substantially, but intensity prediction remains more difficult: recent intensity guidance has reduced errors [9], yet official forecasts in the western North Pacific continue to carry substantial uncertainty [10]. A central reason is that intensity depends on air–sea enthalpy and momentum exchange, upper-ocean heat content, barrier layers, mesoscale eddies, and TC-induced currents [11, 12].

Existing observing platforms each cover only part of this problem. Spaceborne microwave sensors provide broad but complementary and incomplete views of the near-core wind field. Active scatterometers infer surface wind from ocean backscatter, but lose sensitivity at high wind speeds and can be degraded by rain-induced attenuation and scattering [13]. Passive radiometers infer wind from ocean emissivity and can penetrate some cloud cover, but cloud liquid water, precipitation emission, and atmospheric attenuation contaminate the brightness-temperature signal [13, 14]. Neither sensor class resolves the subsurface ocean [15]. Aircraft obtain high-quality near-core measurements but are expensive and campaign-limited [16, 17], whereas gliders remain at sea for long periods but move much more slowly than a translating TC [18]. AUVs can adapt their sampling to changing environmental and vehicle states [19]. Upper-ocean observations can improve initial states and, in some cases, intensity forecasts [20, 21, 22, 23, 24], but their value is highly sensitive to sampling location and background structure. Targeted observation studies likewise show that forecast benefit depends on sensitive regions, the background state, and the sampling design [25, 26, 27]. These limitations leave near-core wind and upper-ocean states poorly constrained in many forecast initial conditions. Adaptive TC observation requires

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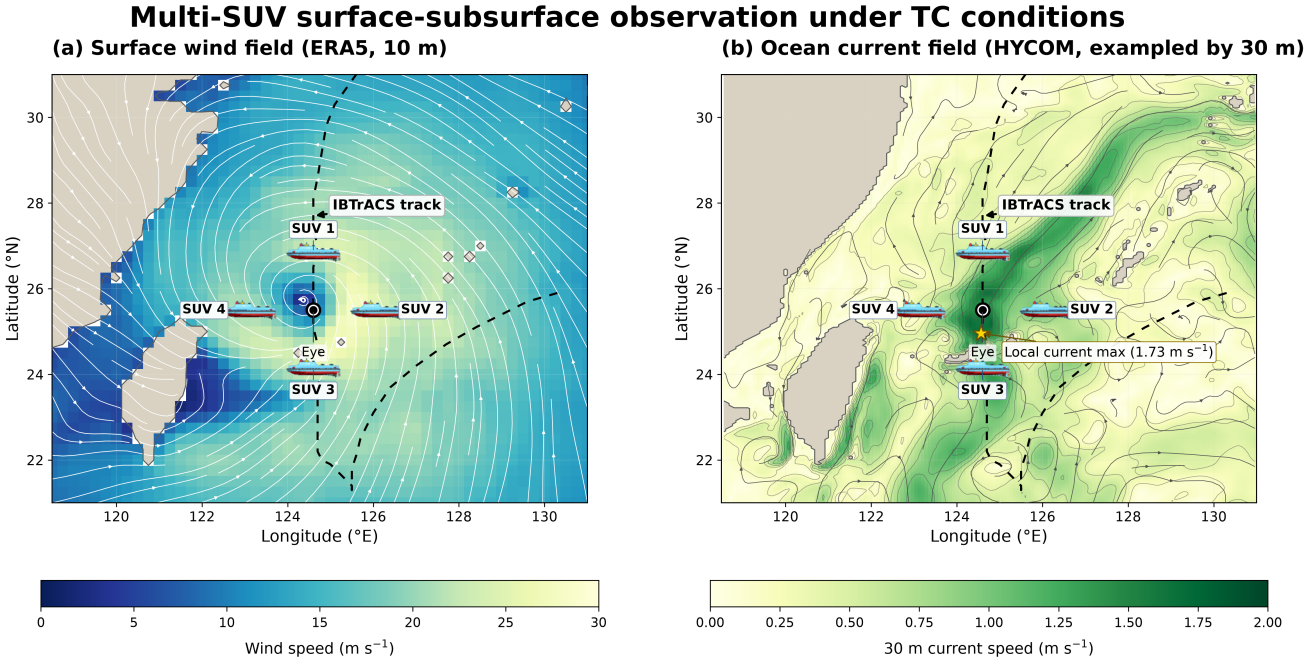


Figure 1: Coordinated surface–subsurface TC observation for a representative HINNAMNOR (2022) snapshot. The left panel shows four SUVs tracking the IBTrACS best track while sampling ERA5 winds at 10 m above the sea surface. The right panel shows the same formation sampling HYCOM currents at 30 m below the sea surface; this depth is shown as an illustrative example, whereas the complete experiments cover the seven depths from 0 to 60 m. The four-vehicle configuration is illustrative, with equal angular spacing; subsequent experiments evaluate fleets of 8, 12, and 16 vehicles.

decisions about motion, sampling time, and coordination, not coverage alone.

A submersible unmanned vessel (SUV) combines rapid surface transit with subsurface sensing [28, 29, 30, 31]. The operational concept assumes that the SUVs are deployed from a mother ship, such as a large research vessel or an equivalent offshore support ship, or from an offshore floating platform such as the Shanghai Jiao Tong University Far-Seas Floating Island (Yuanhai Fudongdao). Such platforms can provide rapid relocation, long-duration station keeping, and deployment of heavy marine equipment [32, 33, 34]. The Blue Whale is used as an engineering reference: its public specifications indicate a surface speed of about 23 kn (11.83 m/s), a submerged speed of about 3 kn (1.54 m/s), and an observational payload that includes an acoustic Doppler current profiler and a CTD sensor, with a public reference depth of approximately 60 m [28]. This study uses the current Blue Whale's surface–subsurface motion capabilities as the reference parameterization and evaluates a possible future-upgrade configuration as a comparison case. The vehicles are simplified as point masses for path planning. In the proposed mission, the surface fleet observes 10 m winds and follows the moving TC core. Final tracking error measures the fleet-center-to-TC-core distance remaining at the end of each surface episode; validation values are summarized by the median across episodes. At the subsurface, four SUVs are assigned to each prescribed layer and seek the strongest observable current. Figure 1 illustrates this surface-to-subsurface task switch; four vehicles are shown at the surface as an example, and the same shared-action surface policy is later evaluated with 4, 8, 12, and 16 SUVs.

Traditional graph search, sampling-based planning, potential fields, and model predictive control are effective when the cost map and dynamics are known [35, 36, 37, 38, 39, 40]. In a TC, however, the fields are nonstationary and only locally observed, while propulsion is superimposed on ambient flow. Conventional pipelines also estimate a field or waypoint sequence before execution. By contrast, reinforcement learning (RL) maps each new measurement directly to the next action, closing the observation–decision–motion loop. RL has demonstrated strong sequential decision making across diverse domains [41, 42, 43, 44, 45, 46, 47, 48], and independent PPO is competitive in cooperative multi-agent tasks [49, 50, 51]. Multi-AUV adaptive sampling and target tracking provide the ocean-robotics foundations [52, 53, 54, 55, 56].

Three gaps remain. First, a unified atmosphere–ocean reinforcement-learning environment for coordinated surface–subsurface TC observation has not been established [57, 58]. Second, it remains unclear whether parameter-sharing MARL can track a moving TC under deployment uncertainty, sample the subsurface current field, and remain effective across different fleet sizes [25, 26, 27, 58]. Third, learned policies have rarely been compared with causal planners under identical local observations, motion constraints, and control interfaces. We formulate stochastic policy optimization under partial observability [59, 49, 50],

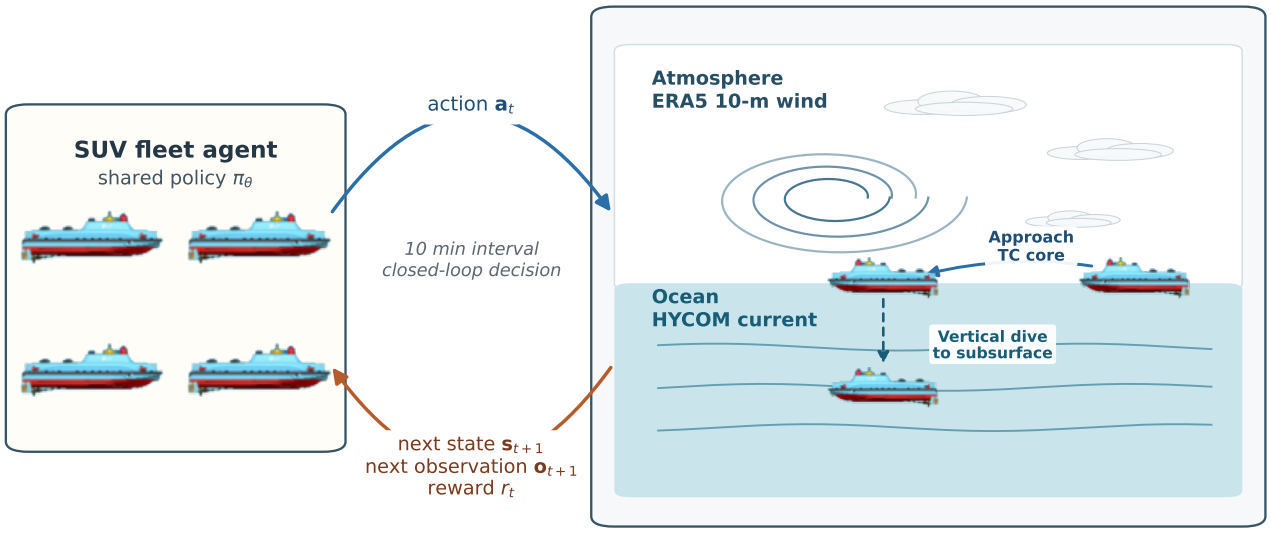


Figure 2: Closed-loop interaction between the SUV fleet agent and the tropical-cyclone atmosphere–ocean environment. At each 10 min interval, the SUV fleet agent perceives environmental information as observations. Its shared policy π_θ maps the common surface observation to a joint formation action. Underwater, each SUV maps its local current-history observation to an independent action. The environment returns a reward and the next state to the agent. In the right panel, the rightmost surface SUV senses the local wind speed and moves toward the TC core region. After reaching a more central position, represented by the left surface SUV, the first-stage task is completed. The SUV then dives vertically and begins the second-stage observation task at the prescribed depth, collecting temperature, salinity, and current observations and intelligently adjusting its motion based on the local current information.

evaluate 4-, 8-, 12-, and 16-SUV fleets, and provide this matched comparison.

The main contributions of this study are summarized as follows.

1. We construct a unified atmosphere–ocean RL environment by aligning ERA5 10 m winds, HYCOM currents at seven depths from 0 to 60 m, and IBTrACS tracks. The final 400-TC dataset is divided at the TC level into 360 training TCs and 40 held-out evaluation TCs.
2. We evaluate an IPPO policy with globally shared surface-wind observations and one joint formation action for 4-, 8-, 12-, and 16-SUV fleets. Subsurface controllers retain decentralized current histories. Surface policies are tested at 100–800 km deployment offsets, and subsurface policies are tested at seven depths and two propulsion limits. Surface tracking statistics use episode medians, whereas subsurface current and coverage statistics use episode means.
3. We compare IPPO with four traditional planners, including artificial potential field, receding-horizon A*, lawnmower coverage, and information-greedy control, under matched local observations, action constraints, motion constraints, control intervals, and identical evaluation episodes.

2. Problem Formulation

2.1. Observation Task and Design Requirements

The objective is to coordinate a fleet of submersible unmanned vessels (SUVs) to observe a moving tropical cyclone (TC) across the sea surface and upper ocean. At the surface, the SUV fleet navigates as a rigid formation toward the moving TC core. Final tracking error measures the fleet-center-to-TC-core distance remaining at the end of the episode and serves as a proxy for access to the near-core high-wind regime. In the operational concept, the fleet would descend after entering a predefined terminal tracking region. The learning experiments evaluate this mission as two separate episodes: a surface approach episode and a subsurface sampling episode in which four SUVs seek the highest attainable current speed at each prescribed depth. The mission imposes four operational requirements:

1. rapid correction of initial deployment errors using surface propulsion;
2. tracking of the moving TC using onboard wind observations and inter-vehicle observation sharing;
3. subsurface observation at ocean depths of 0, 10, 20, 30, 40, 50, and 60 m below the sea surface;
4. safe operation under action smoothing and coastline safeguards.

The surface maximum speed is varied between 12 and 20 m/s, while the submerged maximum propulsion speed is

1.5 or 2.0 m/s. This difference creates an asymmetric decision problem: surface propulsion is used to follow the TC and reduce the fleet-center-to-TC-core distance, quantified primarily by the final tracking error. At the subsurface, the primary learning objective changes to finding and sampling higher-current water, while geometric and current-weighted spatial coverage characterize how broadly and how valuably the fleet samples the dynamically active region. The four SUVs must balance active propulsion against passive advection by the ambient current.

2.2. Problem Definition

The fleet contains N SUVs, where $N \in \{4, 8, 12, 16\}$, indexed by $i \in \{1, 2, \dots, N\}$. The position, commanded horizontal velocity, and operating mode of SUV i at time t are denoted by $\mathbf{x}_i(t)$, $\mathbf{p}_i(t)$, and $m_i(t) \in \{\mathrm{s}, \mathrm{u}\}$, respectively, where s and u denote the surface and subsurface modes. Horizontal positions and reported distances are expressed in km, whereas commanded and ocean-current velocities are expressed in m/s. The commanded speed satisfies

$$\|\mathbf{p}_i(t)\| \le \begin{cases} V_{\max}^{\mathrm{s}}, & m_i(t) = \mathrm{s}, \\ V_{\max}^{\mathrm{u}}, & m_i(t) = \mathrm{u}. \end{cases} \tag{1}$$

Each simulated episode is assigned to one stage, consistent with the separate surface and subsurface environments used in training and evaluation. The mode $m_i(t)$ selects the applicable speed bound and motion law for that episode; continuous diving, ballast transitions, and resurfacing are not simulated within a single decision trajectory.

For the motion equations, let $\mathbf{p}_i$ denote the desired horizontal velocity command and $\mathbf{q}_i^{\mathrm{prop}}$ the effective propulsion velocity after the acceleration and heading dynamics. The km/min representation of the ocean current is denoted by $\mathbf{v}_{\mathrm{ocean}}$; the device-level speed bounds above remain reported in m/s. The surface uses the effective propulsion velocity directly as ground velocity, whereas the underwater stage adds current advection. The horizontal motion in km is

$$\dot{\mathbf{x}}_i(t) = \begin{cases} \mathbf{q}_i^{\mathrm{prop}}(t), & m_i(t) = \mathrm{s}, \\ \mathbf{q}_i^{\mathrm{prop}}(t) + \mathbf{v}_{\mathrm{ocean}}\left(\mathbf{x}_i(t), z_i, t\right), & m_i(t) = \mathrm{u}, \end{cases} \tag{2}$$

where $z_i \in \{0, 10, 20, 30, 40, 50, 60\}$ m. Time t is measured in minutes, and $\mathbf{v}_{\mathrm{ocean}}(\cdot)$ is the HYCOM current at the corresponding position and depth. Underwater ground velocity is the superposition of active propulsion and background advection.

For numerical implementation, the continuous model is integrated explicitly at the decision interval $\Delta t = 10$ min:

$$\mathbf{x}_{i,k+1} = \mathbf{x}_{i,k} + \Delta t \begin{cases} \mathbf{q}_{i,k}^{\mathrm{prop}}, & m_{i,k} = \mathrm{s}, \\ \mathbf{q}_{i,k}^{\mathrm{prop}} + \mathbf{v}_{\mathrm{ocean}}(\mathbf{x}_{i,k}, z_i, t_k), & m_{i,k} = \mathrm{u}. \end{cases} \tag{3}$$

Here $\Delta t = 10$ min. The current field is evaluated at the beginning of each interval and interpolated to the current vehicle position. For the active underwater reward, we write this pre-move current as $\mathbf{u}_{i,k} = \mathbf{u}_{\mathrm{ocean}}(\mathbf{x}_{i,k}, z_i, t_k)$.

To represent the dynamic constraints of the SUV, the desired command is converted into a speed and course and then tracked under bounded acceleration, deceleration, heading rate, and minimum turn radius. In the surface stage, the command specifies a desired horizontal ground velocity, which is tracked without an added current term. In the underwater stage, the command specifies a desired propulsion velocity before current advection is applied:

$$\begin{aligned} \mathbf{p}_{i,k}^{\mathrm{des}} &= V_{\max}^{m_i} \tilde{\mathbf{a}}_{i,k}, \qquad s_{i,k}^{\mathrm{des}} = \left\| \mathbf{p}_{i,k}^{\mathrm{des}} \right\|_2, \\ \psi_{i,k}^{\mathrm{des}} &= \operatorname{atan2}\left(\bar{a}_{i,k,y}, \bar{a}_{i,k,x}\right). \end{aligned} \tag{4}$$

Each SUV carries an effective propulsion speed s_i and heading ψ_i, with $\mathbf{q}_i^{\mathrm{prop}} = s_i[\cos\psi_i, \sin\psi_i]^\top$. The 10 min control interval is divided into ten 60 s substeps. In substep ℓ, speed and heading evolve according to

$$\begin{aligned} s_i^{\ell+1} &= \mathrm{clip}\big(s_i^{\ell} \\ &\quad + \mathrm{clip}\Big(s_i^{\mathrm{des}} - s_i^{\ell}, -a_{m_i}^{\mathrm{dec}}\Delta\tau, a_{m_i}^{\mathrm{acc}}\Delta\tau\Big), 0, V_{\max}^{m_i}\Big), \\ \bar{\omega}_i^{\ell+1} &= \min\left(\omega_{\max}^{m_i}, \frac{s_i^{\ell+1}}{R_{\min}^{m_i}}\right), \\ \psi_i^{\ell+1} &= \mathrm{wrap}\big[\psi_i^{\ell} \\ &\quad + \mathrm{clip}\big(\mathrm{wrap}\left(\psi_i^{\mathrm{des}} - \psi_i^{\ell}\right), -\bar{\omega}_i^{\ell+1}\Delta\tau, \bar{\omega}_i^{\ell+1}\Delta\tau\big)\big], \end{aligned} \tag{5}$$

where $\Delta\tau = 60$ s. For the surface stage, $a_{\mathrm{s}}^{\mathrm{acc}} = 0.02$ m/s^2, $a_{\mathrm{s}}^{\mathrm{dec}} = 0.03$ m/s^2, $\omega_{\max}^{\mathrm{s}} = 1°$/s, and $R_{\min}^{\mathrm{s}} = 200$ m. For the underwater stage, $a_{\mathrm{u}}^{\mathrm{acc}} = 0.002$ m/s^2, $a_{\mathrm{u}}^{\mathrm{dec}} = 0.003$ m/s^2, $\omega_{\max}^{\mathrm{u}} = 2°$/s, and $R_{\min}^{\mathrm{u}} = 100$ m. The minimum-radius term is applied only while the vehicle is moving; a commanded zero action decelerates the vehicle without changing its course. The vehicle then advances over each substep using

$$\mathbf{x} \leftarrow \mathbf{x} + \left(\mathbf{q}_i^{\mathrm{prop}} + \mathbf{1}_{\{m_i = \mathrm{u}\}}\mathbf{u}_{\mathrm{ocean}}\right)\Delta\tau, \tag{6}$$

with the current held fixed over the 10 min control interval. Thus the surface desired command is a ground velocity, whereas the underwater desired command is a propulsion velocity before current advection. The speed is initialized to zero and the heading is initialized uniformly in $[0, 2\pi)$ at the beginning of each episode.

The motion model remains a horizontal point mass model and does not resolve added mass, hydrodynamic drag, roll and pitch, buoyancy and ballast dynamics, thruster transients, wave loads, or actuator delay. ERA5 winds are available at 0.25° and hourly resolution, whereas HYCOM currents are available at approximately 0.04° by 0.08° and three-hourly resolution. These fields are interpolated to the 10 min decision interval and to each vehicle position, but such interpolation does not recover high-frequency wave forcing, local wake effects, or unobserved vehicle attitude

and actuator states. A detailed six-degree-of-freedom model would require platform-specific hydrodynamic coefficients and forcing information that are not commensurate with the present data, and could imply a level of numerical precision unsupported by the inputs. The results therefore address planning and coordination rather than structural loading or low-level controller response.

2.3. Surface–Subsurface Two-Stage Task

The SUVs first perform surface tracking, using coordinated propulsion to follow the TC and reduce their distance to its center. For the operational protocol, we assume that descent begins after the surface fleet enters the terminal region defined by a final tracking error of at most 200 km. The simulation uses this 200 km value as the upper bound on the tracking error at the surface-to-subsurface transition and initializes the subsurface vehicles within the corresponding 200 km disk around the TC core at the end time of the surface segment. Because the present simulation does not model descent dynamics, the surface and subsurface stages are represented as two time-aligned but state-independent episodes. The descent duration is simplified: after the surface stage, each vehicle is assumed to arrive immediately at one prescribed depth. Each depth is a separate configuration with four SUVs, and a single episode does not visit multiple depths. At the end of the underwater observation task, the operational concept returns the SUVs to the sea surface for recovery. This ascent and recovery phase is not modeled here; the paper models only the surface observation-and-tracking stage and the subsequent fixed-depth underwater observation stage.

The local observation of SUV i is

$$\mathbf{o}_i(t) = \begin{cases} \left[\mathbf{s}_i(t),\, \mathbf{v}_i(t)\right], & m_i(t) = \mathrm{s}, \\ \left[\mathbf{h}_i(t),\, \mathbf{v}_i(t)\right], & m_i(t) = \mathrm{u}, \end{cases} \tag{7}$$

where $\mathbf{s}_i(t)$ is the stage-specific measurement block, $\mathbf{v}_i(t)$ is the normalized effective horizontal propulsion velocity $\mathbf{v}_i = \mathbf{q}_i^{\mathrm{prop}}/V_{\max}^{m_i(t)}$, and $\mathbf{h}_i(t)$ is an optional causal history block. In the final surface experiments, $\mathbf{s}_i(t) \in \mathbb{R}^{3N}$ contains the normalized speed and two wind components at 10 m above the sea surface sampled by all N SUVs; the surface policy has no history block, and its effective ground velocity is augmented by $[\cos\psi_i, \sin\psi_i]$. Surface operation assumes idealized, lossless, and instantaneous inter-vehicle communication: every SUV receives the wind observations acquired by all other SUVs in the fleet as part of its policy input. In the underwater IPPO evaluation, no inter-vehicle communication is assumed. The observation contains no separate instantaneous-current block in addition to the history; instead, define the local current feature $\mathbf{c}_{i,t} = [\|\mathbf{u}_{\mathrm{ocean}}(\mathbf{x}_{i,t}, z_i, t)\|, u_{\mathrm{ocean},x}, u_{\mathrm{ocean},y}]^\top$, where $\mathbf{u}_{\mathrm{ocean}} = (u_{\mathrm{ocean},x}, u_{\mathrm{ocean},y})$, and $\mathbf{h}_i(t) = [\mathbf{c}_{i,t-4}^\top, \ldots, \mathbf{c}_{i,t}^\top]^\top \in \mathbb{R}^{15}$, which is the causal five-frame history of SUV i's current speed and two components at its assigned depth; before five measurements are available, the earliest observed frame is repeated to fill the history. Each underwater SUV receives neither the current observations nor the positions or actions of the other SUVs. The complete reanalysis field, the global best-track center, and future environmental states are not provided to the policy. These two communication assumptions are intentionally different and mark the bounds of the present study. The idealized global surface link represents a favorable satellite or radio-connected operating mode, which makes it an upper-bound assumption. The no-communication underwater case is the conservative opposite bound: shared policy parameters are loaded separately on each vehicle, but no observation, position, or action is exchanged at runtime. The surface payload is one shared vector of $3N$ scalar wind values per fleet decision, whereas the underwater inter-vehicle message payload is zero. The reported experiments do not model bandwidth, packet loss, acoustic latency, or intermittent surfacing. We do not present either the ideal surface link or the fully decentralized subsurface case as a validated communication system. The resulting surface observation dimension is $d_o^{\mathrm{s}} = 3N+4$. The underwater five-frame history contains 15 scalars; adding the two normalized propulsion components and the heading representation $[\cos\psi_i, \sin\psi_i]$ gives $d_o^{\mathrm{u}} = 19$.

During training, the reward function can access the distance between each SUV and the best-track TC core. This distance supervises the association between local 10 m wind patterns and proximity to the TC core/high-wind region; the center coordinate itself is not an actor input. During validation, rewards do not influence actions. The trained policy receives only the shared fleet-level wind samples and must track an unseen TC, thereby testing whether the learned wind-to-center relation generalizes.

The normalized action is

$$\mathbf{a}_i(t) \in [-1, 1]^2, \tag{8}$$

and is projected onto the unit disk before being mapped to a desired velocity command in both modes:

$$\tilde{\mathbf{a}}_i(t) = \frac{\mathbf{a}_i(t)}{\max\left(1, \|\mathbf{a}_i(t)\|_2\right)}, \qquad \mathbf{p}_i(t) = V_{\max}^{m_i(t)} \tilde{\mathbf{a}}_i(t). \tag{9}$$

For the surface task, the hard formation constraint is imposed by sharing both the observation and the action:

$$\mathbf{o}_{i,t} = \mathbf{o}_t, \qquad \mathbf{a}_{i,t} = \mathbf{a}_t, \qquad i = 1, \ldots, N. \tag{10}$$

The formation center is $\bar{\mathbf{x}}_{e,t} = N^{-1} \sum_i \mathbf{x}_{i,e,t}$, and the initial relative offsets $\mathbf{r}_i = \mathbf{x}_{i,e,0} - \bar{\mathbf{x}}_{e,0}$ are held constant. Because the surface model excludes water currents and all SUVs receive the same dynamic command, each vehicle undergoes the same translation while preserving its initial ring, line, grid, or V-shaped geometry. For the subsurface task, the action remains vehicle-specific because each SUV observes only its own current history. This mapping ensures that the desired and effective speeds never exceed the mode-specific speed limit. Here $\bar{\mathbf{a}}_i(t)$ is the executed normalized command after action smoothing, unit-disk projection, and any stage-specific safety projection; without a safety intervention, $\bar{\mathbf{a}}_i(t) = \tilde{\mathbf{a}}_i(t)$. In the surface stage, the action

specifies a desired ground velocity. In the underwater stage, the same unit-disk command specifies a desired propulsion speed and course. In both stages, the effective speed and heading evolve according to the dynamics defined above.

2.4. Cooperative Action Formulation

The cooperative task uses a shared-action partially observable Markov decision process at the surface and a decentralized partially observable Markov decision process (Dec-POMDP) at the subsurface:

$$\mathcal{M} = \langle \mathcal{I}, \mathcal{S}, \{\mathcal{A}_i\}, P, R, \{\Omega_i\}, O, \gamma \rangle, \tag{11}$$

where $\mathcal{I}$ is the set of SUVs, $\mathcal{S}$ is the joint environment state, $\mathcal{A}_i$ and Ω_i are the local action and observation spaces of vehicle i, $P(s_{t+1} \mid s_t, \mathbf{a}_t)$ is the transition probability, $R(s_t, \mathbf{a}_t, s_{t+1}) \in \mathbb{R}^N$ is the local-reward vector, and $O(\mathbf{o}_{t+1} \mid s_{t+1}, \mathbf{a}_t)$ is the joint observation function. The shared team reward before clipping is $\bar{r}_t = N^{-1} \sum_{i=1}^{N} r_{i,t}$, where $r_{i,t}$ is the ith element of $R(s_t, \mathbf{a}_t, s_{t+1})$, and γ is the discount factor. For the surface stage, $\mathcal{A}_i = \mathcal{A}$ and $\Omega_i = \Omega$ for every SUV, and the joint action is constrained to $\mathbf{a}_{1,t} = \cdots = \mathbf{a}_{N,t} = \mathbf{a}_t$. The resulting process is a centralized partially observable control problem for the rigid formation. For the subsurface stage, the action and observation spaces remain vehicle-specific as in a conventional Dec-POMDP.

At each 10 min interval, the policy forms a closed perception–decision–action loop and does not follow a precomputed route:

$$\begin{aligned} &\text{surface: } \mathbf{o}_t \to \mathbf{a}_t = \pi_\theta(\mathbf{o}_t) \to \{\mathbf{x}_{i,t+1}\}_{i=1}^{N} \to \mathbf{o}_{t+1}, \\ &\text{subsurface: } \mathbf{o}_{i,t} \to \mathbf{a}_{i,t} = \pi_\theta(\mathbf{o}_{i,t}) \to \mathbf{x}_{i,t+1} \to \mathbf{o}_{i,t+1}. \end{aligned} \tag{12}$$

The updated observation becomes the next control input. This feedback structure lets the fleet adapt to changing wind and current fields, partial observability, and deployment uncertainty without a static waypoint sequence. Fig. 2 summarizes this interaction between the moving atmosphere–ocean environment and the cooperative SUV fleet.

3. Data and Simulation Environment

3.1. Reanalysis and Best-Track Products

ERA5 supplies hourly zonal and meridional wind components at 10 m above the sea surface on a regular 0.25° grid [1] [60]. Wind speed and direction are calculated from u_{10} and v_{10}. HYCOM supplies three-hourly zonal and meridional currents at ocean depths of 0, 10, 20, 30, 40, 50, and 60 m below the sea surface [2] [61]. The native horizontal grid is approximately 0.04° in latitude by 0.08° in longitude. IBTrACS provides TC time, center position, maximum wind, minimum pressure, and related metadata at mostly 3 or 6 h intervals [3] [62]. The dataset-selection domain used to identify candidate systems is 98.7°E–171.6°E and 6.8°N–56.7°N. The TC events span October 2007 to December 2025 UTC. The surface observation variable is the 10 m wind speed and direction, whereas the subsurface observation variables are the horizontal current velocity, represented by `water_u` and `water_v`, and the derived current speed over the 0–60 m layer. Table 1 summarizes the products, variables, native grids, and temporal resolutions used to construct the simulation environments.

3.2. Spatiotemporal Alignment and Track Segmentation

Hourly ERA5 winds and three-hourly HYCOM currents are linearly interpolated to a common 10 min simulation clock and queried at each SUV position. The environmental trajectories are precomputed and do not respond to vehicle motion; this one-way forcing assumption is appropriate for planning at the scale of a TC. The policy receives only local reconstructed values at the current time, not future fields or the complete reanalysis map. The best-track center is used only for training rewards and evaluation.

Commands are held constant within each decision interval:

$$\mathbf{a}_i(t) = \mathbf{a}_{i,k}, \quad t_k \le t < t_{k+1}, \quad t_{k+1} - t_k = \Delta t = 10\ \text{min}. \tag{13}$$

The environmental fields, TC track, action, and position update use the same time grid. The 10 min step balances control responsiveness against the hourly and three-hourly resolution of the source products; a shorter step would add samples without adding resolved information. The same interval is used for IPPO and all traditional planners.

Each IBTrACS track is first linearly interpolated to the 10 min simulation interval. The cumulative along-track distance is computed using the local equirectangular approximation

$$\Delta s_k = \sqrt{\left[111 \Delta\lambda_k \cos(\varphi_k)\right]^2 + \left(111 \Delta\varphi_k\right)^2}, \tag{14}$$

where $\Delta\lambda_k$ and $\Delta\varphi_k$ are longitudinal and latitudinal increments in degrees, and 111 km is the mean meridional arc length per degree. Target segment lengths are sampled uniformly from 300–500 km, with a fixed 250 km stride, 50–250 km overlap, and an average overlap of about 150 km. Residual track shorter than 300 km is discarded. After assignment to 10 min track samples, the final realized lengths range from 292.1 to 505.5 km, with a median of 396.1 km.

The IBTrACS archive contains 540 tropical cyclones in the study domain between October 2007 and December 2025 UTC. We retain TCs with aligned ERA5 and HYCOM products and apply two data-quality screens. First, for each candidate segment, every 10 min TC position must remain at least 200 km from land. Second, we remove segments that

[1] ERA5 download portal: https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels.

[2] HYCOM GLBy0.08 dataset: https://www.hycom.org/data/glby0p08/expt-93pt0; HYCOM ESPC-D-V02 dataset: https://www.hycom.org/data/espcd-v02.

[3] IBTrACS download portal: https://www.ncei.noaa.gov/products/international-best-track-archive.

Table 1
Environmental datasets used to construct the 400-TC observation environments. The dataset integrates ERA5 surface winds, HYCOM currents at seven depths, and IBTrACS best tracks.

Data product	Source	Variables used	Native grid	Native time
Surface wind at 10 m above sea level	ERA5	u_{10}, v_{10}, speed, direction	$0.25° \times 0.25°$	Hourly
Current at 0–60 m ocean depth	HYCOM	water_u, water_v, speed at 0/10/20/30/40/50/60 m	$\sim 0.04°$ lat $\times 0.08°$ lon	3-hourly
Best track	IBTrACS	Position, time, wind, pressure	Point track	Mainly 3/6-hourly

Note: although the SUV platform can carry sensors for observing temperature, salinity, and currents, the reinforcement-learning policy uses only surface wind and subsurface current observations as environmental inputs.

start at the TC formation boundary ($t_0 = 0$) and segments that extend beyond the final valid IBTrACS timestamp; the latter prevents constant-position extrapolation at the end of a track. The resulting 400-TC dataset contains 3,392 segments. The 360 training TCs contribute 3,130 segments, whereas the 40 held-out TCs contribute 262 segments. The split is made at the TC level, so no TC appears in both sets. Segment durations range from 2.2 to 97.5 h. The median duration was 15.3 h and the mean was 17.7 h. These segments provide mission-scale evaluation windows. The platform can carry current, CTD, pH, chlorophyll, and turbidity sensors [28], while the control experiments use current velocity as the decision-making observation.

4. Multi-Agent Reinforcement Learning Method

4.1. Blue Whale-Class Platform Modeling

Each SUV is represented by a horizontal point mass model with mode-dependent speed bounds and position updates from Eqs. (1)–(2). The surface actor uses globally shared 10 m wind observations and outputs one common velocity command for the rigid formation, whereas the underwater actor uses a causal five-frame history of its local current and normalized velocity. The surface control variable is defined in the Earth-fixed frame as a desired ground velocity. Therefore, ocean-current compensation is not included as a separate term in the surface position update. Underwater motion retains advection because 1.5–2.0 m/s propulsion is comparable to the TC-induced current.

The two simulation configurations compare present and potential future platform capabilities. The 12 m/s surface and 1.5 m/s submerged limits are close to the public Blue Whale design speeds, whereas 20 m/s and 2.0 m/s represent a future-upgrade case. Table 2 reports the complete parameterization and distinguishes modeling assumptions from certified specifications.

Across 28,506 IBTrACS track samples, the median, 95th-percentile, and 99th-percentile TC translation speeds are 5.14, 13.89, and 20.58 m/s (Table A.1). The 12 and 20 m/s surface limits cover 93.0% and 98.8% of these samples, respectively. The submerged limits cover only 3.3% and 7.0% of the samples. For the subsurface task, the SUVs are initialized randomly within a 200 km radius of the TC core and perform local adaptive sampling; the task does not attempt to overtake or continuously follow the moving TC core.

4.2. Parameter-Sharing Independent PPO

All SUVs contribute to the same tracking objective, and the training return is defined as the team-average reward. At the surface, the cooperative task is represented by a single joint formation action for the whole fleet. At the subsurface, each SUV retains an independent action and local observation.

For the surface task, IPPO optimizes one two-dimensional joint formation action from the shared fleet observation, and every SUV executes this same action. The fixed relative offsets in Eq. (10) constitute a hard constraint rather than a reward penalty. For the subsurface task, IPPO uses parameter sharing with vehicle-specific observations and action samples. The main surface model uses a memoryless multilayer perceptron, whereas the active underwater controller uses a five-frame current history. The shared cooperative return is the team-average reward. The surface value estimate is $V_\phi(\mathbf{o}_t) = v_\phi(\mathbf{o}_t)$, whereas the subsurface estimate sums the local values, $V_\phi(\mathbf{o}_t) = \sum_{i\in\mathcal{I}} v_\phi(\mathbf{o}_{i,t})$. A single clipped PPO objective is optimized over the surface formation action or the factorized multi-agent subsurface action:

$$L^{\mathrm{CLIP}}(\theta) = \mathbb{E}_t\Big[\min\Big(\rho_t(\theta)\hat{A}_t, \; \mathrm{clip}\left(\rho_t(\theta), 1-\epsilon, 1+\epsilon\right)\hat{A}_t\Big)\Big], \tag{15}$$

where

$$\rho_t(\theta) = \prod_{i\in\mathcal{I}} \frac{\pi_\theta(a_{i,t} \mid o_{i,t})}{\pi_{\theta_{\mathrm{old}}}(a_{i,t} \mid o_{i,t})}, \tag{16}$$

and $\hat{A}_t$ is estimated by generalized advantage estimation from the team-average reward. The actor objective is complemented by a clipped value loss and an entropy bonus. For the surface stage, the product in Eq. (16) reduces to the single shared-action likelihood. For the subsurface stage, it retains the vehicle-specific likelihoods. During surface execution, one policy query produces $\mathbf{a}_t$, which is transmitted unchanged to every SUV. Underwater, each SUV computes its own action from its local observation and the shared policy network. The complete training and deployment procedure is summarized in Algorithm 1.

To compare reinforcement-learning algorithm choices, IPPO, SAC, and TD3 were trained for the same 2×10^6

Table 2
Public Blue Whale reference and the two simulation configurations. The second simulation column is a simple parameterization of the present-day reference platform, whereas the third represents a possible future-upgrade configuration with higher surface and subsurface speeds for comparison. These scenarios are modeling assumptions, not certified vehicle data.

Property	Public reference	Current-platform scenario	Future-upgrade scenario
Surface speed	23 kn, ≈11.83 m/s	12 m/s	20 m/s
Submerged speed	3 kn, ≈1.54 m/s	1.5 m/s	2.0 m/s
Public-reference submerged endurance	≥4 h at 3 kn; ≥72 h standby	–	–
Underwater evaluation horizon	–	20 h simulation	20 h simulation
Sampling depth	≈60 m	0–60 m, seven layers	0–60 m, seven layers
Payload	≥800 kg	800 kg	800 kg
Surface fleet size	Single platform	4/8/12/16	4/8/12/16
Decision interval	Not specified	10 min	10 min
Surface formation	–	100 km ring, $360°/N$	100 km ring, $360°/N$

Algorithm 1 Training and deployment of shared-action surface IPPO and independent subsurface IPPO

Input: Training TC split, stage-specific environment, fleet size N, speed limit $V_{\max}$, reward coefficients, and random seed
Output: Shared actor and critic parameters

Initialize the actor, critic, and parallel environment workers
for training iteration $m = 1, \dots, M$ **do**
  Reset the stage-specific environment and obtain the common surface observation $\mathbf{o}_t$, or the local observations $\mathbf{o}_{i,t}$ for all $i = 1, \dots, N$ underwater
  for rollout step t **do**
    if surface stage **then**
      Sample or select one joint formation action $\mathbf{a}_t \sim \pi_\theta(\cdot \mid \mathbf{o}_t)$ as the joint formation action for all SUVs
    else
      Sample or select a local action for each SUV from $\pi_\theta(\cdot \mid \mathbf{o}_{i,t})$
    end
    Apply the stage-specific action filter, unit-disk projection, and common formation safety filtering
    Advance all SUVs and collect the team-average reward
  end for
  Compute GAE advantages and returns for the team-average reward
  Update the shared actor and critic with the clipped PPO objective
end for
Freeze the actor and load the same parameters on every SUV
for each deployment episode **do**
  Observe local or shared data according to the stage communication model
  Compute the shared surface action, or the local subsurface action, from the policy
  Apply action smoothing, circular projection, and safety filtering
  Execute the action and observe the next state
end for

environment steps on representative surface and subsurface cases with two independent training seeds. Surface metrics are summarized by episode medians and underwater metrics by episode means before averaging across the two seeds. This comparison is used only as bounded algorithm-choice evidence. IPPO requires no replay buffer and is therefore used for the main, formation, and planner comparisons.

The complete reward coefficients, safety parameters, network dimensions, and PPO hyperparameters are summarized in Appendix C.

4.3. Surface and Subsurface Reward Design

The surface objective is to follow the translating TC as closely as possible and reach the high-wind region, with the final tracking error as the primary operational metric. Let $d_{i,k+1}$ denote the post-transition distance from SUV i to the TC core after executing the command at decision step k, and let $\bar{\mathbf{a}}_{i,k}$ denote the executed command after filtering and unit-disk projection. The surface reward used in the

implementation is (19)

$$r_i^{\mathrm{surf}} = -\frac{\min(d_{i,k+1}, D_{\max})}{D_{\mathrm{scale}}} + b_{\mathrm{near}}\mathbf{1}_{d_{i,k+1}<d_{\mathrm{near}}} - \frac{W_{\mathrm{form}}\max(0, d_{i,k+1} - R_{\mathrm{form}})}{D_{\mathrm{scale}}} - \lambda_u\|\bar{\mathbf{a}}_{i,k} - \bar{\mathbf{a}}_{i,k-1}\|_2^2 + r_i^{\mathrm{coast}}, \tag{17}$$

where the first term is a bounded proximity cost and the second term provides a discrete near-center bonus with $b_{\mathrm{near}} = 1$. Here $D_{\max} = 300$ km caps the proximity cost, $D_{\mathrm{scale}} = 100$ km normalizes the distance terms, and $d_{\mathrm{near}} = 100$ km defines the near-center reward region. The third term adds an additional radius-excess penalty beyond $R_{\mathrm{form}} = 100$ km, and the fourth term penalizes abrupt action changes with $\lambda_u = 0.05$. The term r_i^{coast} represents the coastal-safety reward defined below. The 300 km cap prevents distant errors from producing an excessively steep gradient and reduces premature policy collapse during early training. After aggregation over SUVs, the team average reward is clipped to $[-5, 2]$.

For subsurface observation, four SUVs are assigned to each prescribed depth. Their primary objective is not to follow the surface TC core directly, but to observe the highest possible current speed. Geometric and current-weighted spatial coverage are then evaluated as complementary measures of observation extent and observation value. Low-speed SUVs cannot generally overtake a translating TC, and the vertical projection of the surface center is not necessarily the location of the strongest subsurface current. For subsurface IPPO training, the current-seeking reward is

$$r_i^{\mathrm{sub}} = \min\left(\frac{\|\mathbf{u}_{i,k}\|}{V_{\mathrm{norm}}}, R_{\max}\right) - \lambda_a\|\tilde{\mathbf{a}}_{i,k}\|_2^2 + \beta_{\|}\max(0, \cos\theta_i) + r_i^{\mathrm{coast}}, \tag{18}$$

where $V_{\mathrm{norm}} = 1.0$ m/s, $R_{\max} = 2.0$, $\lambda_a = 0.05$, and $\beta_{\|} = 0.5$. The term $-\lambda_a\|\tilde{\mathbf{a}}_{i,k}\|_2^2$ penalizes large normalized propulsion commands and regularizes the action magnitude. The angle θ_i denotes the direction difference between the effective propulsion velocity and the pre-move current $\mathbf{u}_{i,k}$, so the positive-alignment term is activated only when the SUV pushes downstream; formally, we define $\cos\theta_i = 0$ when either vector is zero and otherwise set $\cos\theta_i = (\mathbf{q}_{i,k}^{\mathrm{prop}}\cdot\mathbf{u}_{i,k})/(\|\mathbf{q}_{i,k}^{\mathrm{prop}}\|_2\|\mathbf{u}_{i,k}\|_2)$. The current-intensity reward is evaluated at $\mathbf{x}_{i,k}$ before the state update.

4.4. Coastal Safety and Action Smoothing

The land mask is converted into a nonnegative offshore-distance field. A vehicle inside land receives a large penalty, and a vehicle within the buffer distance D_{buf} receives

$$r_i^{\mathrm{coast}} = \begin{cases} -P_{\mathrm{land}}, & D_i = 0, \\ -W_b\max(0, 1 - D_i/D_{\mathrm{buf}}), & 0 < D_i < D_{\mathrm{buf}}, \\ 0, & D_i \geq D_{\mathrm{buf}}, \end{cases}$$

where $D_i = D(\mathbf{x}_{i,k+1})$ is the post-transition offshore distance, with $D_i = 0$ on land, and P_{land} and W_b are positive penalty coefficients.

For the surface stage, three complementary mechanisms reduce coastline intrusion: offshore-only rejection sampling during deployment, a soft nearshore penalty, and an action-level safety filter. The main surface experiments use a minimum deployment distance of 100 km, a 100 km nearshore buffer, and a 100 km safety margin. Best-of-rejection deployment favors offshore initial states, the reward discourages trajectories near land, and the safety filter removes the onshore action component when an SUV approaches the shoreline. For the subsurface stage, the nearshore penalty is retained and an episode is truncated at landfall, but offshore-only initialization and action-level safety filtering are not applied.

To reduce control oscillations and thereby lower mean speed and shorten path length, the main surface experiments use an exponential action filter:

$$\mathbf{a}_k^{\mathrm{filt}} = (1 - \beta)\mathbf{a}_k^{\mathrm{raw}} + \beta\mathbf{a}_{k-1}^{\mathrm{filt}}, \tag{20}$$

where $\beta \in [0, 1)$ controls the degree of temporal smoothing. The filter uses only the current and previous actions, which prevents future-information leakage. The filtered command $\mathbf{a}_k^{\mathrm{filt}}$ is then passed through the unit-disk projection and speed mapping in Eq. (9).

4.5. Surface and Subsurface Evaluation

The center-distance reward is used only during training and is absent from validation. Validation tests whether a trained policy can infer the TC direction from wind observations in unseen events. The primary surface metric is the final tracking error, defined for each validation episode as the fleet-center-to-TC-core distance at the final time step. A smaller value places the fleet closer to the TC core, where near-core winds are generally stronger. The mean tracking error is the episode-level time average of the fleet-center-to-TC-core distance and is reported as a secondary process diagnostic. For both indicators, the reported validation statistic is the median across validation episodes rather than the arithmetic mean. Subsurface metrics instead use the arithmetic mean over validation episodes within each trained policy seed; the reported uncertainty is the sample standard deviation of those seed-level means.

For episode e at decision step k, the formation center is

$$\bar{\mathbf{x}}_{e,k} = \frac{1}{N}\sum_{i=1}^{N}\mathbf{x}_{i,e,k}. \tag{21}$$

For episode e, the episode-level mean tracking error is

$$d_e^{\mathrm{avg}} = \frac{1}{K_e}\sum_{k=1}^{K_e}\left\|\bar{\mathbf{x}}_{e,k} - \mathbf{x}_{\mathrm{TC},e,k}\right\|_2, \tag{22}$$

where K_e is the number of post-transition decision intervals. For training seed s, the reported final and mean tracking

errors are the medians over validation episodes:

$$D^{(s)}_{\text{final}} = \underset{e=1,\dots,M_s}{\text{median}} \left\| \bar{\mathbf{x}}_e(T_e) - \mathbf{x}_{\text{TC},e}(T_e) \right\|_2 , \tag{23}$$

$$D^{(s)}_{\text{mean}} = \underset{e=1,\dots,M_s}{\text{median}} \, d_e^{\text{avg}}, \tag{24}$$

where T_e is the final decision time of validation episode e, M_s is the number of validation episodes, and distance is computed from the formation center rather than from each SUV independently. The reported value is the arithmetic mean of the seed-level medians. The traditional planners are deterministic and have no training seed. Their reported statistics are therefore medians over all paired validation episodes; different deployment realizations are treated as validation episodes, not as trained-model seeds.

The median is used because episode-level tracking errors are strongly right-skewed. A small number of episodes can produce terminal errors far above the typical case, particularly when another active TC produces a competing wind signature or when the local wind field provides a weak and unstable centre-directed gradient. An arithmetic mean would let these rare cases dominate the reported central tendency even though they do not represent nominal operation. These cases remain in the evaluation and are discussed explicitly in Section 5.1 and in the limitations, rather than being silently removed from the evaluation set.

Surface evaluation covers 4/8/12/16 SUVs on a 100 km ring with equal angular spacing. The formation-center offset represents the random position error of a TC track forecast: it is equivalent to deploying the fleet at the corresponding distance from the true TC core with a randomly selected bearing, after which the SUVs move toward the TC. During training, formation-center offsets are sampled uniformly within an 800 km radius, covering deployment-error scenarios from 0 to 800 km. During evaluation, the trained policies are tested at fixed radii of 100, 200, 400, 600, and 800 km to quantify how the magnitude of the forecast-induced deployment error affects tracking performance. Metrics are first summarized by the median over validation episodes within each seed and then averaged across independent training seeds.

For the subsurface evaluation, each depth from 0, 10, 20, 30, 40, 50, and 60 m is tested as a separate 20 h configuration with four SUVs and 1.5 or 2.0 m/s speed limits. We assume that, after tracking under the 0–800 km random surface deployment errors, the fleet reaches a core region within 200 km of the TC core and begins descent. Underwater performance is evaluated under this condition. Because descent time and vehicle-by-vehicle endpoint transfer are not modeled, each SUV is initialized directly at its assigned depth, with the formation center sampled uniformly within a 0–200 km radius of the TC core. Thus, 200 km represents the current upper bound of the residual random position error of the formation center relative to the TC core at the surface-to-subsurface transition, rather than a fixed operating limit. As SUV performance improves, this core-region threshold could be tightened to 100 or 50 km; the present experiments use 200 km. There is no inter-vehicle communication, so each SUV uses only its own five-frame current history and propulsion state. IPPO and the traditional planners are evaluated on the same episodes, deployment seeds, motion model, and speed bound. The IPPO policy uses its deterministic mean action during evaluation.

For underwater evaluation, we consider three metrics: mean experienced current speed, geometric coverage, and current-weighted coverage. The first metric measures the current intensity encountered along each SUV's trajectory. Let K be the number of recorded position increments, let $k = 0, \dots, K$ index the archived positions, and let z denote the observation depth. The time-averaged experienced current speed of SUV i in episode e is

$$\bar{V}_{i,e}(z) = \frac{1}{K_e + 1} \sum_{k=0}^{K_e} \left\| \mathbf{u}_{\text{ocean}}(\mathbf{x}_{i,k}, z, t_k) \right\|_2 . \tag{25}$$

where only valid ocean positions are included; a terminal on-land position is excluded from the current and coverage statistics. The reported fleet value is the average $N^{-1} \sum_{i=1}^{N} \bar{V}_{i,e}(z)$. A higher value indicates that the SUVs spent more of their trajectories in stronger-current regions, but this metric does not quantify how broadly the fleet sampled the surrounding area.

Geometric coverage measures the unique ocean area sampled by the fleet in episode e:

$$C_e(z) = A_{\text{cell}} \left| \bigcup_{i=1}^{N} \bigcup_{k=0}^{K_e} \mathcal{G}(\mathbf{x}_{i,k}) \right|, \tag{26}$$

where $\mathcal{G}(\mathbf{x})$ is the 0.25° grid cell containing position $\mathbf{x}$, using the lower-left corner of the aligned event domain as the grid origin; $|\cdot|$ counts unique cells, and $A_{\text{cell}} \approx (0.25° \times 111 \text{ km})^2 \approx 770.06 \text{ km}^2$ is the nominal grid-cell area used for cross-method comparison. A larger $C_e(z)$ indicates a broader spatial sampling footprint, while repeatedly visiting the same cell does not increase this metric. Coverage alone, however, does not distinguish whether the sampled cells contain strong or weak currents.

To represent both current intensity and spatial extent, we define the episode-level current-weighted coverage

$$C_{\text{w},e}(z) = \left[\frac{1}{N} \sum_{i=1}^{N} \bar{V}_{i,e}(z) \right] C_e(z), \tag{27}$$

and aggregate it by taking the arithmetic mean over validation episodes within each training seed, followed by the mean and sample standard deviation of the seed-level means. This quantity assigns a higher value when strong-current observations are combined with a broad sampling footprint. It therefore captures observation value more directly than either the mean experienced current or geometric coverage alone.

For all three underwater metrics, values are first averaged over the validation episodes within each training seed and then averaged across the ten independent seeds.

4.6. Traditional Planning Baselines

To provide a non-learned reference for the learned cooperative policy, we next introduce four traditional planning baselines. We include four non-learned planners as secondary references [36, 38, 40]. The artificial-potential-field planner moves along the local gradient of observed wind or current speed and adds an offshore repulsion inside the coastal buffer. The receding-horizon A* planner builds an online local value map, re-plans a three-step minimum-cost route, executes the first action, and refreshes the plan every 15 decisions. The boustrophedon lawnmower planner follows fixed parallel coverage legs with phase offsets between vehicles and uses no environmental value map. The information-greedy planner evaluates eight candidate unit actions and selects the one with the largest predicted local wind or current value plus the coastal preference.

None of the planners receives the TC core, the complete reanalysis field, future frames, or the learned policy state. Surface planners use the same shared wind samples and fleet navigation state as IPPO. Underwater, each vehicle maintains a private map because communication is disabled. All four use the same 10 min interval, speed envelope, circular projection, surface safety filter. They require no training and have no training-seed variance. Surface planners are evaluated on all 262 held-out segments. Underwater planners use identical sets of 400 evaluation episodes per depth-speed setting. These episodes comprise 40 test TCs with ten deployment seeds each.

5. Simulation and Results

5.1. Surface Active Tracking

Using the evaluation protocol in Section 4.5, we evaluate the trained surface policies across four fleet sizes, two speed limits, five deployment radii, and ten independent training seeds. Final tracking error in Eq. (23) is the primary metric. Mean tracking error is a secondary diagnostic of how closely the fleet remains to the moving TC, while path length and mean ground speed characterize motion demand.

Figure 4 reports the population-level speed-limit comparison. Averaged over fleet sizes and training seeds, the 20 m/s policy obtains final tracking errors of 204.8, 217.2, 272.5, 379.3, and 540.8 km at deployment radii of 100, 200, 400, 600, and 800 km. The corresponding 12 m/s values are 236.7, 255.1, 335.5, 468.4, and 651.3 km. Thus, 20 m/s gives the lower final tracking error in all 20 matched fleet-size/radius comparisons, and its fleet-averaged advantage grows from 31.9 km at 100 km to 110.5 km at 800 km. Both policies remain farther from the center as deployment error increases, because finite speed and episode duration cannot remove every initial displacement.

The mean tracking error shows the same ordering: 20 m/s has the smaller distance in all 20 matched comparisons, with a median 12-minus-20 m/s difference of 49.7 km on identical evaluation episodes. The two-sided tests on identical evaluation episodes give $p < 0.05$ in every case; these tests are exploratory because they are not corrected across the 20 comparisons. The final-tracking-error benefit is larger than the mean-tracking-error benefit at both ends of the deployment range, indicating that the higher-speed policy continues reducing residual separation late in the episode.

The 12 m/s policy reduces path length per SUV by 83.4–220.1 km across the five radii and lowers mean ground speed by 1.75–3.93 m/s, whereas the 20 m/s policy achieves closer final tracking.

The representative trajectories clarify how the aggregate result is produced. Fig. 3 follows the four-SUV, 400 km KHANUN case. The vehicles first turn along the observed wind structure and then reduce their separation from the center; by 19.5 h they have entered the high-wind branch. The hard formation constraint keeps the relative offsets fixed, so the individual paths are parallel translations of the same formation trajectory.

Figure A.1 separates the effects of fleet size and deployment error. At 100–200 km, the formations converge rapidly and stay concentrated near the TC core. At 600–800 km, trajectories lengthen and fan out before turning inward, and final positions remain farther from the center. Increasing the fleet from 4 to 16 SUVs adds sample branches and broadens the approach region, but it does not replace the long-range transit. In the representative 12- and 16-SUV cases, paths overlap near the TC core once the shared wind observations guide the fleet toward the same high-wind region. Deployment error therefore controls the required transit distance, whereas fleet size mainly controls how broadly the approach is sampled.

Two retained long-tail episodes illustrate why the median is preferred for the main statistics. The first is the 2012 MARIA–PRAPIROON dual-TC case. Although MARIA is the designated target, PRAPIROON is simultaneously present in the same ERA5 wind field. The competing TC produces a stronger and more coherent local wind signature along part of the route, and the shared-observation policy moves into the corresponding branch, ending farther from the designated MARIA centre. This case reflects ambiguity in a multi-TC environment rather than the nominal single-target behaviour. The second is a 2021 OMAIS episode at a 200 km deployment radius. Around the fleet centre, the mean local wind speed is only about 4.2 m/s and the mean horizontal wind-speed gradient is approximately 0.008 $\mathrm{m\,s^{-1}\,km^{-1}}$. Such a weak gradient does not provide a persistent centre-directed signal, so the policy can maintain an incorrect heading for a prolonged period. These episodes are retained in the evaluation and are revisited as limitations below; because they occupy the far tail, the median better represents typical performance than the arithmetic mean.

5.2. Surface Formation Comparison

The ring, line, grid, and V-shaped formations provide four different ways to allocate spatial observations across eight SUVs. The V formation places four SUVs symmetrically on each arm. To make the comparison independent of total formation size, all four layouts are normalized to the

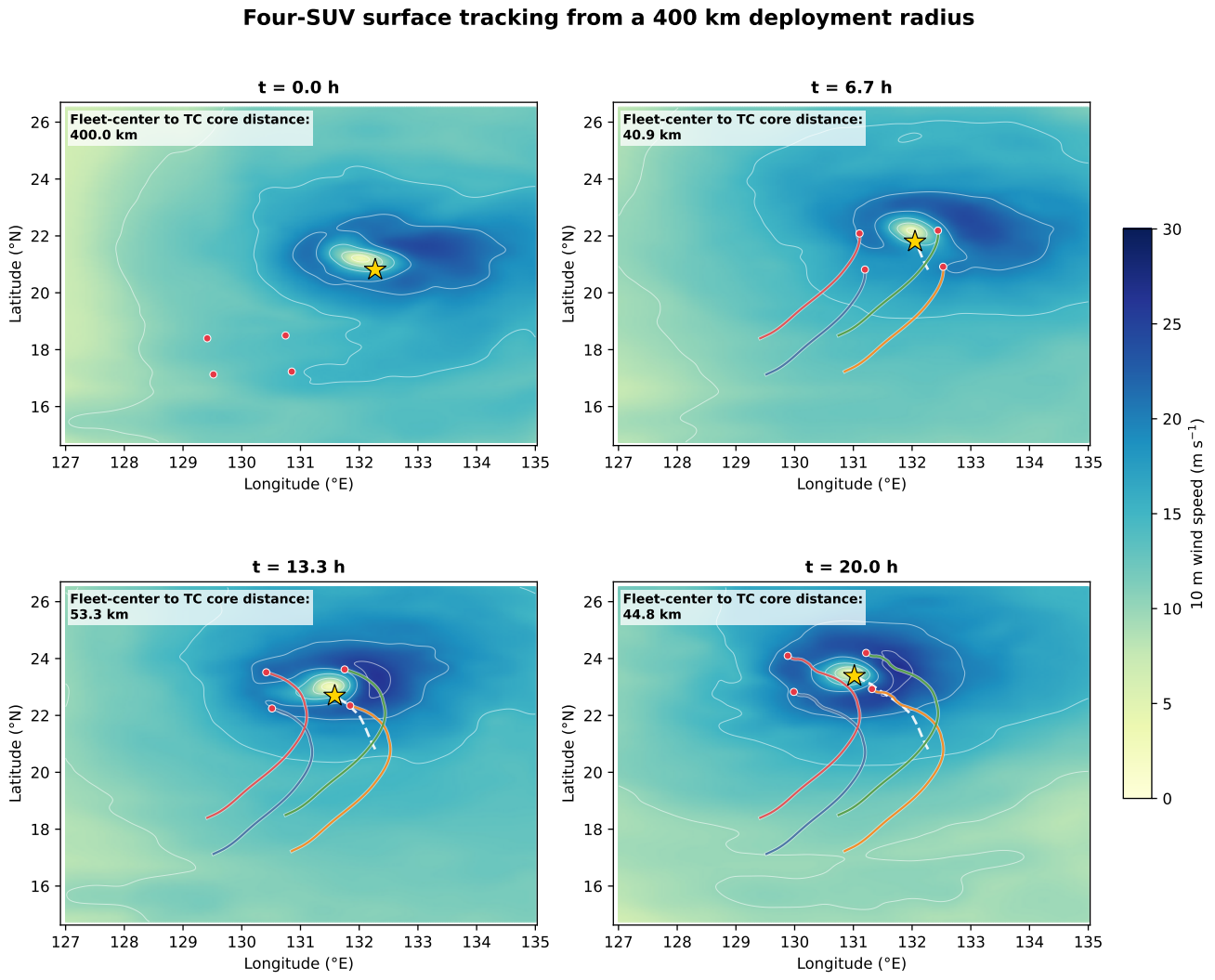


Figure 3: Four-SUV surface-tracking process for KHANUN (2023) at the 20 m/s speed limit. The four panels show the initial deployment and subsequent states at approximately 6.5, 13.0, and 19.5 h. The color background is the spatially interpolated ERA5 10 m wind-speed field at each time. The formation center is deployed at a fixed distance of 400 km from the TC core using a randomly selected bearing. The fleet continually adjusts its trajectory toward the TC core, and the displayed fleet-center-to-TC-core distance decreases from 400.0 km at deployment to 44.8 km at the end of the episode. Small red dots mark vehicle positions, and the yellow star marks the TC core.

same mean radius of 100 km. This common mean radius keeps the fleet spatially spread out and allows it to acquire broader and more numerous coverage observations. This experiment is a matched single-seed sensitivity study: all four policies use training seed 0, the same 2-million-step protocol, deployment distribution, and safety layer at both 12 and 20 m/s, and are evaluated without retraining at 100–800 km (Table A.2). Averaged over the ten speed–radius conditions, the line gives the lowest mean final tracking error of 320.8 km, followed by the V formation at 323.1 km, the ring at 341.2 km, and the grid at 396.0 km. The line is best in six speed–radius conditions, and the V formation is best in the remaining four conditions at 100–600 km under 12 m/s. The ring is best at 800 km under 12 m/s. The V formation also lowers the mean tracking error to 330.8 km, compared with 332.5 km for the line, 348.4 km for the ring, and 377.4 km for the grid.

The mean radius is 100 km for every formation. It is a prescribed geometric property of the initial layout and does not change during an episode under the hard formation constraint.

The grid produces the shortest per-vehicle path, averaging 392.7 km, followed by the ring at 416.5 km, the line at 482.6 km, and the V formation at 499.8 km. The path differences show that the initial arrangement changes the transit effort even though each formation remains rigid after deployment.

Figure 5 visualizes one matched KHANUN segment. Each policy issues one joint formation action, so the initial

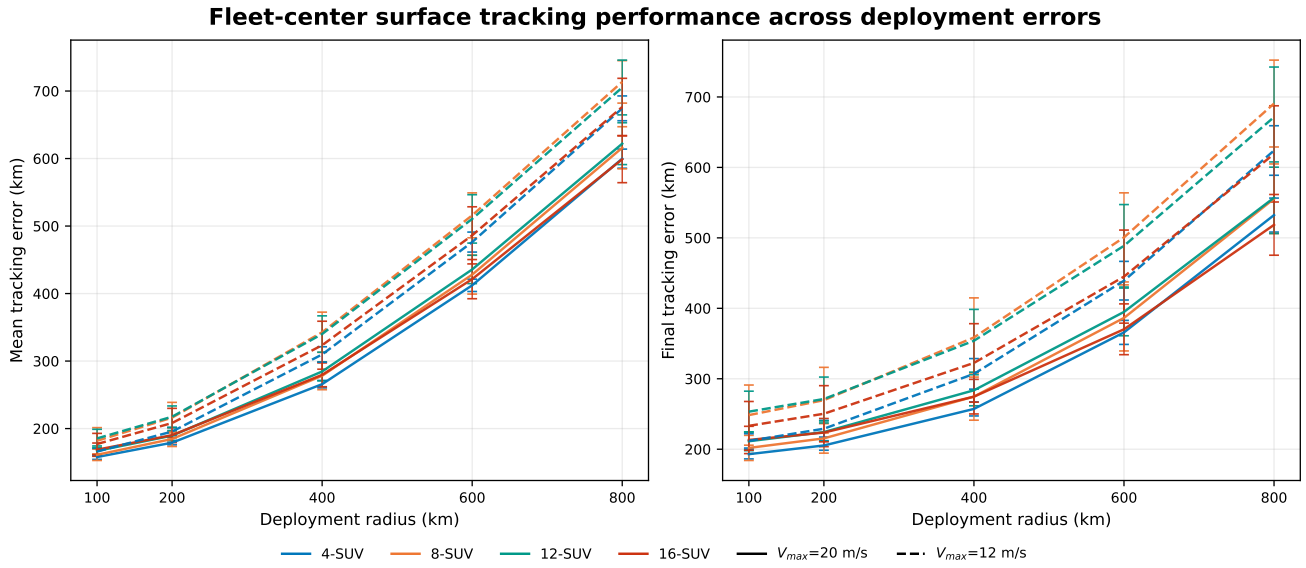


Figure 4: SUV surface tracking performance across deployment errors. The left panel shows the mean tracking error, defined as the mean fleet-center-to-TC-core distance throughout an episode. The right panel shows the final tracking error, defined as the fleet-center-to-TC-core distance at the end of the episode; smaller values indicate a closer fleet center to the TC core. Colors denote fleet sizes and line styles denote the 20 and 12 m/s speed limits. Each point summarizes ten independent training runs using ten random seeds. Error bars show the sample standard deviation.

geometry is preserved throughout the episode. This representative episode visualizes how the same joint formation action produces different spatial trajectories from the four initial arrangements. Because the deployment bearing relative to the TC core is random, the policy cannot assume that the cyclone lies in a fixed direction. The ring formation uses $360°/N$ equal angular spacing and provides direction-independent, approximately isotropic spatial sampling around the formation center. At the same mean radius, the line and V formations concentrate observations along limited directions, whereas the grid concentrates them at a discrete set of locations, especially its corners. We therefore use the ring as the robust default for the main experiments, without claiming that it is optimal for every tracking or path metric. The line, V, and grid formations are reported as sensitivity cases and reflect different emphasis on tracking error and path length. Because only training seed 0 is used, these formation-specific differences are not interpreted as statistically resolved population effects; the ten-seed analysis is reserved for the main ring configuration.

5.3. Execution-Layer Action Filtering

In preliminary rollouts, the unfiltered commands changed direction frequently, causing the SUVs to turn repeatedly and accumulate avoidable path length and propulsion changes. Fig. 6 illustrates this behavior: the unfiltered trajectories form sawtooth-like detours, whereas filtering produces smoother paths under the same policy and initial positions. We therefore added an execution-layer low-pass filter that smooths successive commands while preserving the 10 min decision interval. Table 3 quantifies this change using the same trained actors over identical validation episodes. Without filtering, the 4-, 8-, 12-, and 16-SUV configurations travel 1019.2–1160.0 km per vehicle, with mean speeds of 14.85–16.45 m/s and maximum speeds of 19.46–19.47 m/s. With the execution-layer filter ($\beta = 0.8$), path length falls to 497.4–670.8 km, mean speed to 8.49–11.01 m/s, and maximum speed to 14.25–17.97 m/s. The final tracking error changes from 355.4 to 251.3 km for four SUVs, from 324.7 to 277.8 km for eight SUVs, from 312.0 to 278.8 km for 12 SUVs, and from 260.2 to 256.1 km for 16 SUVs. Averaged over the four fleet sizes, it decreases from 313.1 to 266.0 km, a reduction of 47.1 km or 15.0%. Filtering improves final tracking for every fleet size.

The same comparison shows reductions for every fleet size: path length decreases by 651.7, 614.9, 521.9, and 459.1 km for 4, 8, 12, and 16 SUVs, respectively, while maximum speed decreases by 5.22, 4.14, 3.50, and 1.49 m/s.

5.4. Subsurface Observation and Comparison with Traditional Planners

Before comparing IPPO with non-learned planners, we first isolate its subsurface performance at each observation depth and propulsion limit. Appendix Table A.3 reports experienced current speed, geometric coverage, and current-weighted coverage after averaging over episodes within each training seed and then over ten seeds.

Mean experienced current decreases from 0.534 m/s at the surface to approximately 0.30 m/s at 60 m under both speed limits. At a fixed depth, the 1.5 and 2.0 m/s policies differ by less than 0.006 m/s in mean experienced current. The faster envelope mainly expands the sampled region: relative to 1.5 m/s, geometric coverage is 1.18–1.34 times

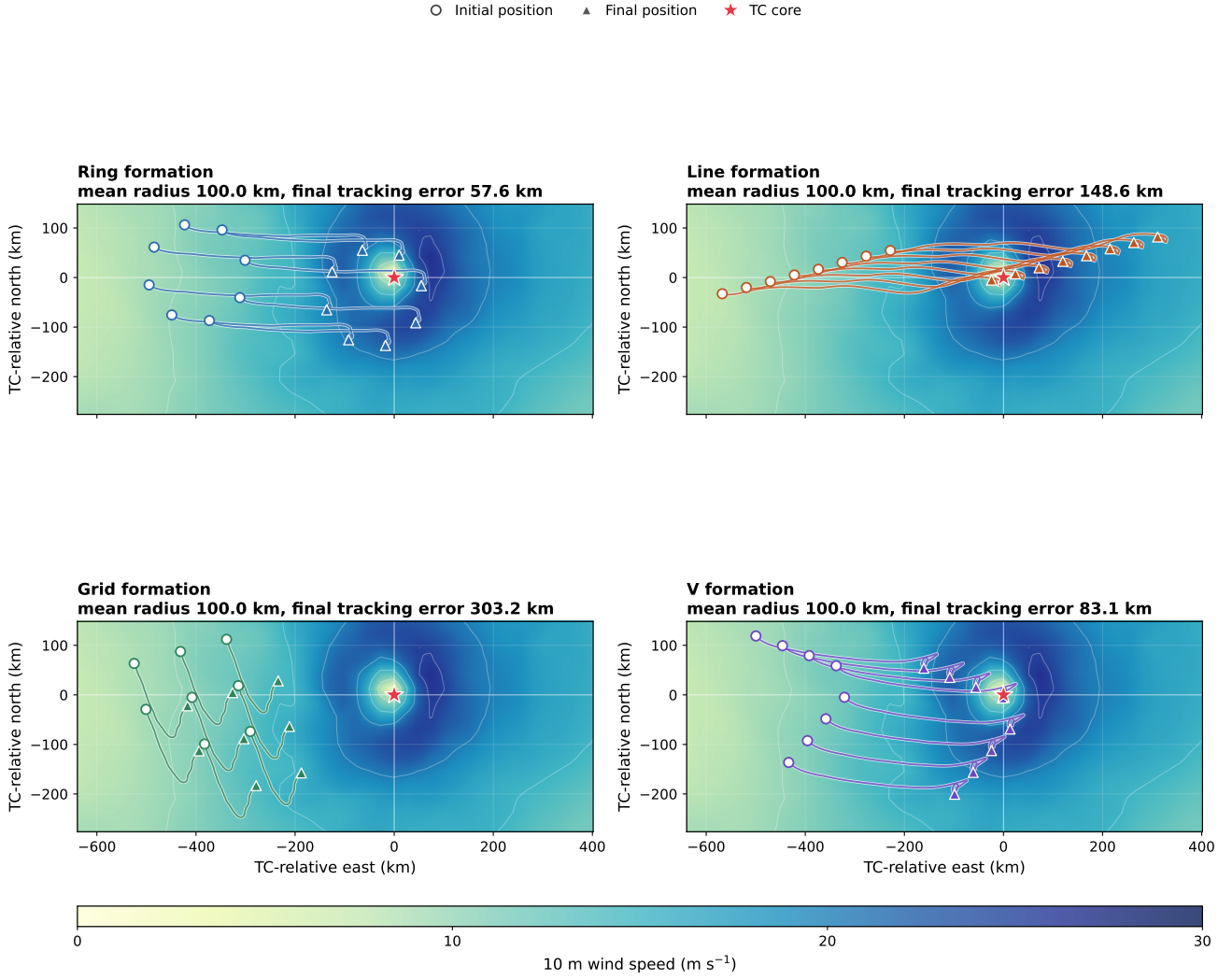


Figure 5: Representative surface trajectory comparison for ring, line, grid, and V-shaped formations with eight SUVs at 12 m/s. All layouts use the same mean radius of 100 km. The formation center is deployed at a distance of 400 km from the TC core using a randomly selected bearing. Open circles denote initial positions, triangles denote final positions, and the red star denotes the TC core. The color background and contours show the spatially interpolated ERA5 10 m wind-speed field at the final simulation step. Axes are relative to the instantaneous TC core.

larger and current-weighted coverage is 1.15–1.30 times larger.

Appendix Figure B.1 shows one matched trajectory example. Table 4 compares IPPO with four traditional planners. Its surface panel provides matched context, whereas the subsurface panel is the focus of this section. Across all 40 surface configurations, IPPO has the lowest final tracking error, mean tracking error, and path length. It produces a lower final tracking error than potential field in 35 of 40 comparisons and than each remaining planner in all 40. Its path is shorter than those of potential field and A* in every comparison and shorter than those of the lawnmower and information-greedy planners in 26 and 34 of 40 comparisons. The path advantage over the latter two planners is therefore less uniform than the final-tracking-error advantage.

For the subsurface task, no planner leads all three metrics. Potential field has the highest mean experienced current at 0.452 m/s. IPPO has the largest geometric coverage at 3506 km^2 and the largest current-weighted coverage $\mathcal{C}_w(z)$. Its mean current of 0.408 m/s is close to that of the lawnmower planner. IPPO's current-weighted coverage exceeds that of potential field, A*, lawnmower control, and information-greedy control in all 14 depth–speed settings by factors of 1.41–1.60, 1.39–1.56, 1.17–1.31, and 1.95–2.28, respectively. Thus, potential field performs better on mean experienced current, whereas IPPO performs better on geometric and current-weighted coverage; no method leads all three metrics.

Appendix Table B.1 confirms the same pattern after stratification. IPPO retains the lowest final tracking error and mean tracking error at every surface deployment radius.

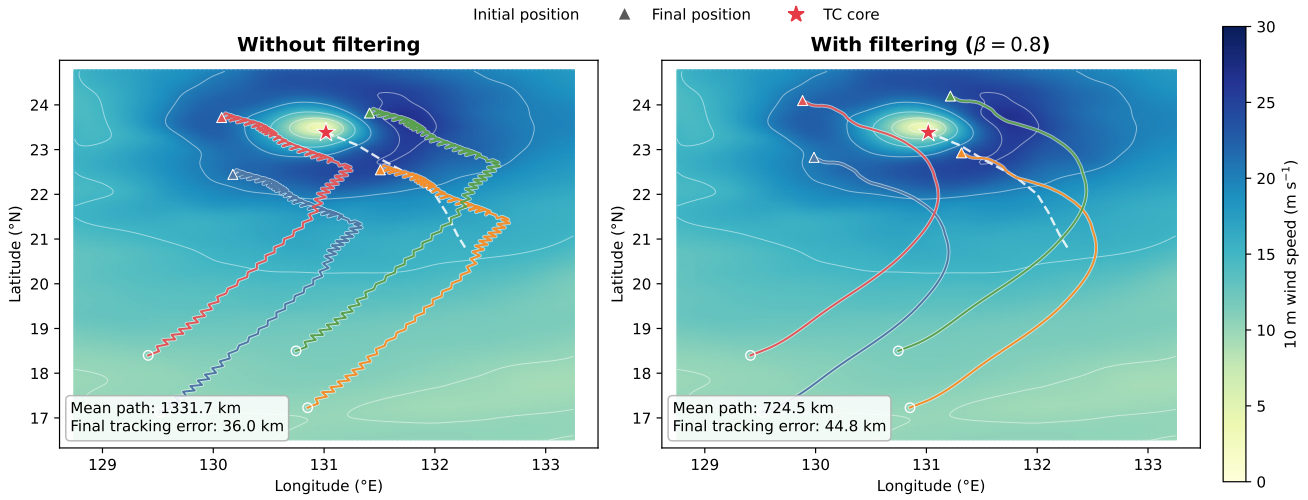


Figure 6: Representative effect of execution-layer filtering on surface paths for four SUVs under the same trained policy, initial positions, deployment radius, and KHANUN (2023) segment. The formation center is deployed at a fixed distance of 400 km from the TC core using a randomly selected bearing. The left panel uses the raw policy command, whereas the right panel applies the low-pass filter with $\beta = 0.8$. Open circles and triangles mark initial and final positions, respectively, and the red star marks the TC core.

Table 3
Execution-layer low-pass filtering for 4/8/12/16 surface SUVs on 262 held-out segments from 40 unseen TCs in the 400-TC dataset. Each trained actor is evaluated with the filter off and on ($\beta = 0.8$) on the same validation episodes. Each entry is the median over validation episodes within a seed, averaged across seeds.

Fleet	Filter	Final tracking error (km)	Path per SUV (km)	Mean speed (m/s)	Maximum speed (m/s)
4	Off	355.4	1160.0	16.45	19.47
4	On	**251.3**	**508.2**	**8.49**	**14.25**
8	Off	324.7	1119.5	16.11	19.47
8	On	**277.8**	**504.7**	**8.65**	**15.32**
12	Off	312.0	1019.2	14.85	19.46
12	On	**278.8**	**497.4**	**8.52**	**15.96**
16	Off	260.2	1129.9	16.29	19.47
16	On	**256.1**	**670.8**	**11.01**	**17.97**

Final tracking error is the fleet-center-to-TC-core distance at the final time step. Off and on use the same trained actor.

Table 4
Comparison with four traditional planning baselines on the held-out TC segments from the same 400-TC train/test split. Surface values are means over the 4/8/12/16-SUV fleets, both speed limits, and all five deployment radii after computing condition medians over validation episodes. Subsurface values use arithmetic means over all seven depths and both propulsion limits. Final and mean tracking errors are measured from the fleet center to the TC core. For surface tracking, lower errors and paths are better. For subsurface observation, higher current and coverage are better.

Panel A: surface tracking

Method	Final tracking error (km)	Mean tracking error (km)	Path per SUV (km)
IPPO	**356.2**	**356.6**	**462.3**
Potential field	413.4	404.2	728.8
Receding-horizon A*	457.2	429.4	823.9
Lawnmower coverage	554.7	478.4	509.1
Information greedy	466.2	429.4	528.7

Panel B: subsurface observation

Method	Mean current (m/s)	Geometric coverage (km^2)	Current-weighted coverage (10^3 $m\,s^{-1}\,km^2$)
IPPO	0.408	**3506**	**1.599**
Potential field	**0.452**	2048	1.050
Receding-horizon A*	0.427	2310	1.097
Lawnmower coverage	0.408	2914	1.286
Information greedy	0.429	1549	0.739

Underwater, potential field has the highest mean current at every depth and IPPO has the largest geometric coverage.

6. Conclusion

This study developed and evaluated a shared-action surface IPPO policy and a parameter-sharing subsurface IPPO policy for coordinated surface–subsurface tropical-cyclone observation with Blue Whale-class SUVs. Current and future-upgrade speed settings were evaluated separately for the surface and subsurface fixed-depth tasks. Historical atmosphere–ocean reanalysis data, including ERA5 and HYCOM, were used to construct the simulation environments for SUV reinforcement-learning training. Training used 360 TCs, and evaluation used 40 unseen TCs.

At the surface, globally shared wind observations allowed the fleet to approach the moving TC despite uncertain deployment. The 20 m/s maximum-speed configuration produced smaller final and mean tracking errors at every deployment radius, and its final-tracking-error advantage grew with deployment error. The 12 m/s maximum-speed configuration produced shorter paths and lower speeds. Action filtering reduced the mean final tracking error at every fleet size and shortened paths and peak speed. In the eight-SUV formation comparison, the line formation had the lowest mean final tracking error, the grid had the shortest paths and widest spacing, and the ring provided a broad sampling footprint and was retained as the main configuration because no formation was uniformly best. In the matched surface comparison with four traditional planners, IPPO had the lowest final and mean tracking errors and generally shorter paths. It had a lower final tracking error than potential field in most matched cases and consistently lower error than the other three planners.

At the subsurface, four SUVs sampled one of seven fixed layers using only their local current histories. The comparison with four traditional planners showed no single winner across all underwater metrics: potential field obtained the highest mean current, while IPPO obtained the largest geometric and current-weighted coverage. Under the same 20 h horizon, increasing the submerged speed limit from 1.5 to 2.0 m/s expanded the sampled area at every depth. A faster future SUV offers greater spatial sampling capacity, provided that endurance and control stability remain feasible.

Nevertheless, several limitations remain. The SUV dynamics are represented by a simplified horizontal point mass model, and calibrated six-degree-of-freedom dynamics are not included. The study targets autonomous environmental sensing and path planning, whereas detailed low-level vehicle control is not modeled. The retained tail cases also expose two scene-level failure modes. In a multi-TC environment, another active cyclone can generate a more coherent local wind signature and divert the fleet from its designated target, as in the 2012 MARIA–PRAPIROON case. In other cases, weak TC winds and a deployment far from the core produce insufficient wind-speed contrast for a stable gradient direction, as in the 2021 OMAIS example. These cases motivate subsequent work on explicit target encoding, active target disambiguation, and wind-observation uncertainty. The main experiments were trained and validated in simulation, and real-vehicle field experiments were not conducted. Future work can incorporate six-degree-of-freedom dynamics and low-level control and validate the algorithms through field experiments in real environments. It should also address the sim-to-real gap in reinforcement learning. In addition, TC wind speeds used as observation inputs to the surface policy may be underestimated in reanalysis datasets, which can affect policy decisions. Future work can use observed TC wind speeds to fine-tune the existing policy through transfer learning.

Code and package availability

The code and processed results are available at `https://github.com/DYHAI/SUV-MARL-TC`. The experiments used Python 3.11, PyTorch 2.11.0 (CUDA 12.8 build), Stable-Baselines3 2.9.0, and Gymnasium 1.2.2 for policy training; NumPy 2.4.6 and pandas 2.3.3 for numerical and tabular processing; SciPy 1.17.1, xarray 2026.7.0, netCDF4 1.7.4, and h5py 3.16.0 for analysis and NetCDF processing; and Matplotlib 3.11.1 for visualization.

Acknowledgements

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A. Supplementary Observation Results

Table A.1: IBTrACS TC translation speeds for the 400-TC dataset. The percentages denote the fraction of main-track samples not exceeding each platform speed. These values describe the speed envelope only; tracking also depends on deployment offset and tracking duration.

Quantity	P50	P90	P95	P99	Samples	TCs
TC translation speed (m s^{-1})	5.14	10.29	13.89	20.58	28,506	400
Envelope	Samples within speed		Samples above speed		Interpretation	
1.5 m/s submerged	3.3%		96.7%		local sampling, not TC chasing	
2.0 m/s submerged	7.0%		93.0%		local sampling, not TC chasing	
12 m/s surface	93.0%		7.0%		current-platform scenario	
20 m/s surface	98.8%		1.2%		future-upgrade scenario	

Table A.2: Surface initial-formation comparison for eight SUVs. All four layouts use a mean radius of 100 km. This is a matched single-seed sensitivity study using training seed 0, with formation-center offsets sampled inside an 800 km radius and evaluation without retraining at fixed offsets of 100, 200, 400, 600, and 800 km on the same 262 held-out TC segments. Final tracking error is the fleet-center-to-TC-core distance at the end of the episode, and path is the per-vehicle trajectory length. Entries are medians over validation episodes. Bold entries are the best value in each speed/radius row; error and path are assessed separately. All distances are in km.

Speed	Train radius	Eval radius	Ring		Line		Grid		V	
(m/s)	(km)	(km)	Final tracking error	Path	Final tracking error	Path	Final tracking error	Path	Final tracking error	Path
12	800	100	210.4	284.4	184.8	330.4	295.8	**219.3**	**165.0**	366.2
12	800	200	228.6	317.2	202.4	364.5	317.7	**241.2**	**183.9**	406.9
12	800	400	312.0	364.2	295.7	409.9	408.5	**272.8**	**284.6**	463.9
12	800	600	445.9	383.9	452.6	408.5	542.6	**291.4**	**438.1**	486.9
12	800	800	**636.5**	372.4	655.6	394.1	733.5	**285.9**	637.5	487.9
20	800	100	197.5	**356.2**	**156.4**	435.3	204.7	383.2	158.1	416.6
20	800	200	209.9	**404.6**	**168.0**	489.5	217.7	433.7	173.8	470.6
20	800	400	266.6	**502.5**	**228.4**	599.5	277.7	537.7	251.8	571.0
20	800	600	371.1	**571.9**	**342.1**	676.6	390.6	616.4	385.3	644.7
20	800	800	533.9	**607.7**	**522.0**	717.2	571.1	645.5	553.3	682.8

Table A.3: IPPO subsurface observation performance at seven depths and two propulsion limits. For every depth–speed setting, values are means over validation episodes within each seed and are then averaged over ten training seeds. Current-weighted coverage is reported in 10^3 m s^{-1} km^2. Bold entries denote the larger value within each depth and metric; equal values remain unbolded.

Depth (m)	V_{max} (m/s)	Mean current (m/s)	Geometric coverage (km^2)	Current-weighted coverage
0	1.5	0.534	3376	2.012
0	2.0	0.534	**3968**	**2.319**
10	1.5	0.472	3385	1.744
10	2.0	**0.476**	**4073**	**2.107**
20	1.5	0.444	3402	1.660
20	2.0	**0.445**	**4037**	**1.956**
30	1.5	**0.409**	3253	1.481
30	2.0	0.408	**3895**	**1.750**
40	1.5	0.364	2851	1.191
40	2.0	**0.368**	**3808**	**1.555**
50	1.5	0.324	2907	1.069
50	2.0	**0.328**	**3749**	**1.374**
60	1.5	0.300	2818	0.959
60	2.0	**0.305**	**3559**	**1.207**

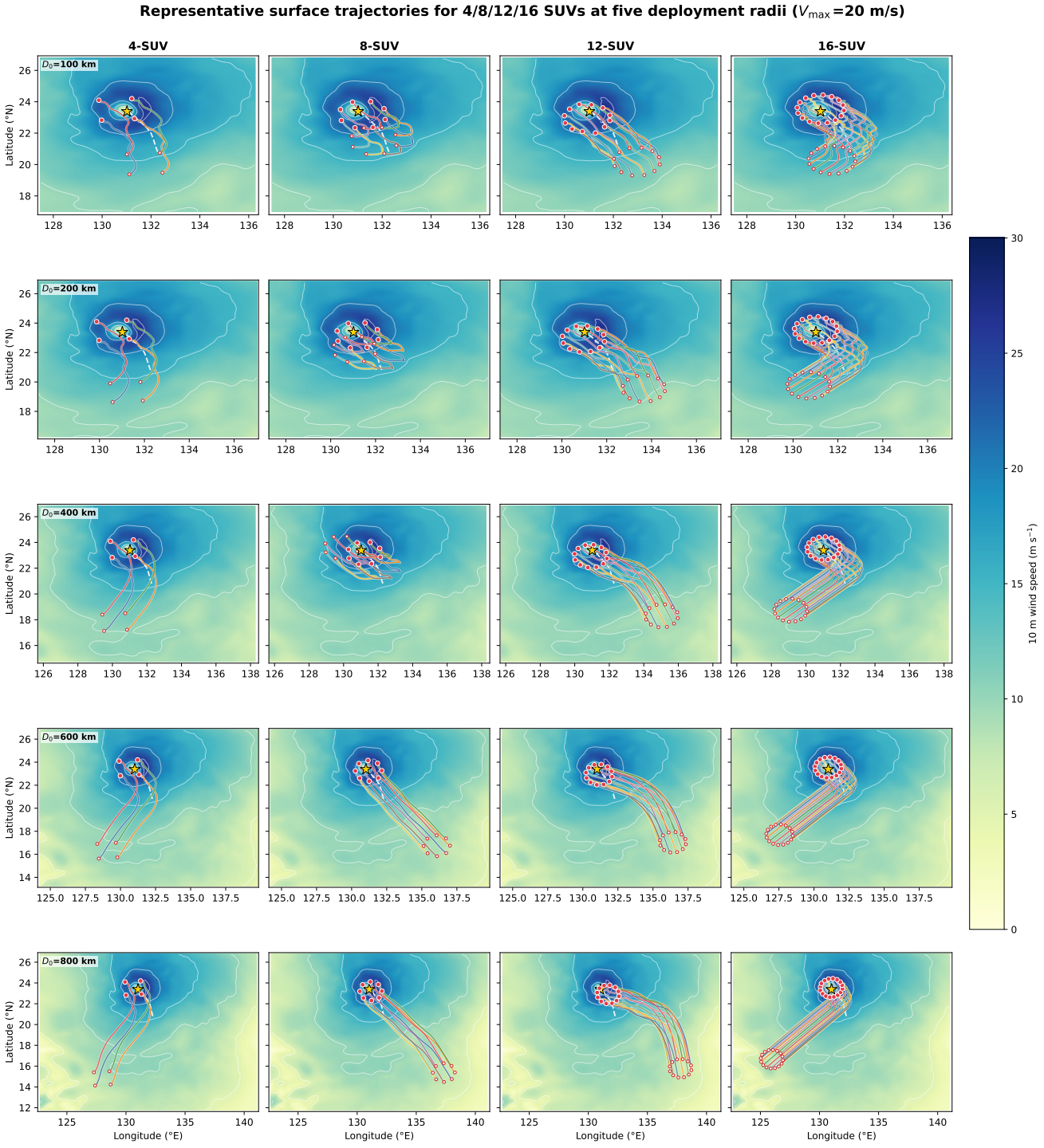


Figure A.1: Representative KHANUN (2023) trajectories for the final 20 m/s surface policy. Columns show 4-, 8-, 12-, and 16-SUV formations, while rows show different formation-center deployment radii of 100, 200, 400, 600, and 800 km. For each radius, the formation center is placed at a randomly selected bearing relative to the TC core. The color background is the spatially interpolated final-time ERA5 10 m wind-speed field. Thin colored lines are vehicle trajectories, open circles mark initial positions, small red dots mark final vehicle positions, and the yellow star marks the TC core at the end of the episode.

B. Traditional Planner Comparison

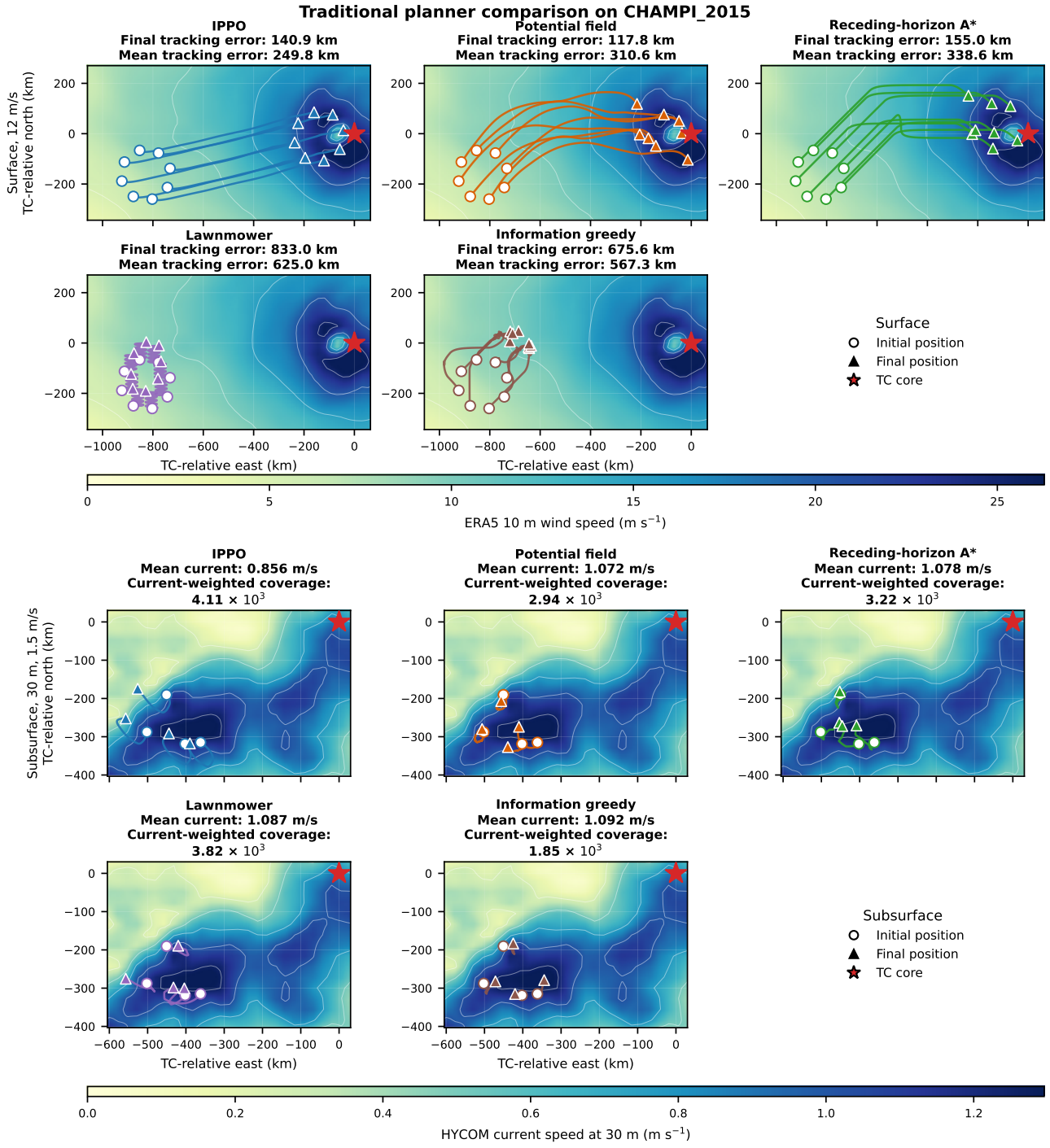


Figure B.1: Comparison between IPPO and the four traditional planners on the same CHAMPI segment. The upper two rows show 12 m/s surface trajectories over the ERA5 10 m wind-speed field. The lower two rows show 1.5 m/s subsurface trajectories at 30 m over the HYCOM current-speed field. All methods use the same initial deployment. The surface formation center is deployed at a distance of 400 km from the TC core, and the subsurface SUVs are initialized within a 200 km radius of the TC core. Panel titles report final and mean tracking errors for the surface task and mean experienced current for the subsurface task. Open circles mark initial positions, triangles mark final positions, and the red star marks the TC core.

Table B.1: Distance and observation metrics for the traditional planner comparison on the held-out TC segments. Panel A reports episode-median final and mean tracking errors by deployment radius, averaged over the four fleet sizes and both speed limits. Panel B reports episode-mean experienced current speed, geometric coverage, and current-weighted coverage at each depth, averaged over the two propulsion limits. Current-weighted coverage is reported in 10^3 m s^{-1} km^2. For surface tracking, lower errors and paths are better. For subsurface observation, higher current and coverage are better.

Panel A: surface tracking metrics						
Radius	Metric	IPPO	PF	A*	Lawn	Greedy
100 km	Final tracking error (km)	**220.8**	290.4	325.8	366.1	295.2
	Mean tracking error (km)	**170.6**	229.9	250.5	232.9	212.0
200 km	Final tracking error (km)	**236.1**	295.4	330.0	394.5	310.7
	Mean tracking error (km)	**197.6**	248.5	267.1	285.9	244.5
400 km	Final tracking error (km)	**304.0**	353.5	394.9	504.5	405.1
	Mean tracking error (km)	**303.0**	343.0	367.4	441.8	374.5
600 km	Final tracking error (km)	**423.9**	476.1	527.6	661.2	560.1
	Mean tracking error (km)	**460.8**	505.1	536.7	621.7	559.1
800 km	Final tracking error (km)	**596.1**	651.8	707.6	847.4	759.8
	Mean tracking error (km)	**650.8**	694.4	725.0	809.7	756.7
Panel B: subsurface metrics by depth						
Depth	Metric	IPPO	PF	A*	Lawn	Greedy
0 m	Mean experienced current (m/s)	0.534	**0.577**	0.548	0.536	0.551
	Geometric coverage (km^2)	**3672**	2214	2445	3067	1579
	Current-weighted coverage	**2.335**	1.443	1.467	1.756	0.956
10 m	Mean experienced current (m/s)	0.474	**0.515**	0.489	0.472	0.492
	Geometric coverage (km^2)	**3729**	2189	2414	3020	1620
	Current-weighted coverage	**1.955**	1.249	1.284	1.513	0.876
20 m	Mean experienced current (m/s)	0.444	**0.486**	0.462	0.444	0.464
	Geometric coverage (km^2)	**3720**	2152	2388	2987	1632
	Current-weighted coverage	**1.877**	1.153	1.204	1.415	0.835
30 m	Mean experienced current (m/s)	0.409	**0.452**	0.429	0.410	0.431
	Geometric coverage (km^2)	**3574**	2105	2329	2939	1588
	Current-weighted coverage	**1.624**	1.039	1.104	1.291	0.754
40 m	Mean experienced current (m/s)	0.366	**0.412**	0.386	0.368	0.391
	Geometric coverage (km^2)	**3330**	1999	2273	2876	1542
	Current-weighted coverage	**1.469**	0.894	0.970	1.141	0.666
50 m	Mean experienced current (m/s)	0.326	**0.374**	0.349	0.327	0.351
	Geometric coverage (km^2)	**3328**	1876	2200	2803	1470
	Current-weighted coverage	**1.246**	0.750	0.848	0.986	0.575
60 m	Mean experienced current (m/s)	0.302	**0.350**	0.325	0.301	0.324
	Geometric coverage (km^2)	**3189**	1799	2120	2702	1410
	Current-weighted coverage	**1.092**	0.658	0.765	0.877	0.519

C. IPPO Algorithm Parameters

Table C.1 summarizes the reward, action-filter, and safety parameters used in the surface and underwater experiments. Table C.2 lists the shared IPPO network and training hyperparameters.

Table C.1: Surface and subsurface reward, action-filter, and coastal-safety parameters used in the final experiments.

Parameter	Symbol	Value	Description and applicable task
Panel A: surface reward and controller parameters			
Proximity-cost distance cap	$D_{\max}$	300 km	Distance cost is capped at 300 km to prevent excessively large gradients from distant initial deployments.
Distance normalization scale	D_{scale}	100 km	Normalization scale used by the proximity and radius-excess reward terms.
Near-center threshold	d_{near}	100 km	Threshold defining the discrete near-center bonus.
Near-center bonus	b_{near}	1.0	Bonus assigned when an SUV is within 100 km of the TC core.
Radius-excess threshold	R_{form}	100 km	Reference radius used by the additional distance penalty beyond 100 km.
Radius-excess weight	W_{form}	1.0	Weight of the penalty applied when an individual SUV is farther than 100 km from the TC core.
Action-change weight	λ_u	0.05	Weight of the squared action change $\|\bar{\mathbf{a}}_{i,k} - \bar{\mathbf{a}}_{i,k-1}\|_2^2$ between successive executed commands.
Action-filter coefficient	β	0.8	Coefficient of $\mathbf{a}_k^{\text{filt}} = (1-\beta)\mathbf{a}_k^{\text{raw}} + \beta\mathbf{a}_{k-1}^{\text{filt}}$. This is a controller setting and is not included in the reward.
Surface reward clipping range	$[r_{\min}, r_{\max}]$	$[-5, 2]$	Clipping range applied to the team-averaged surface reward.
Panel B: underwater IPPO parameters			
Current-speed normalization	V_{norm}	1.0 m/s	Current-intensity reward is implemented as $\min(\|\mathbf{u}\|/V_{\text{norm}}, R_{\max})$.
Current-reward cap	$R_{\max}$	2.0	Upper bound of the normalized current-speed reward.
Downstream-alignment weight	$\beta_{\parallel}$	0.5	Weight of the positive propulsion–current alignment term.
Normalized action-magnitude weight	λ_a	0.05	Penalizes the squared magnitude of the normalized propulsion command. This term regularizes the controller response.
Underwater reward clipping range	$[r_{\min}, r_{\max}]$	$[-5, 5]$	Clipping range applied to the team-averaged underwater reward during IPPO training.
Panel C: coastal-safety parameters by stage			
Land penalty	P_{land}	20	Penalty applied when an SUV enters a land cell in either stage.
Nearshore buffer distance	D_{buf}	100 km	Width of the soft nearshore penalty buffer in both stages.
Nearshore penalty weight	W_b	8.0	Maximum soft-penalty weight within the 100 km nearshore buffer in either stage.
Minimum deployment offshore distance	D_{deploy}	100 km	Minimum coastline distance required by offshore-only rejection sampling during surface deployment; it is not applied to subsurface initialization.
Action-filter safety margin	D_{filter}	100 km	Surface-stage distance below which the onshore component of the commanded action is removed. The subsurface comparison does not use this filter.

Table C.2: Shared-action and parameter-sharing IPPO network, training, and deployment hyperparameters used in the final surface and subsurface experiments.

Parameter	Symbol	Value	Description
Algorithm	–	PPO	Stable-Baselines3 PPO with one shared surface formation action and independent per-agent action sampling underwater.
Optimizer	–	Adam	Adam optimizer used for the actor and critic parameters.
Policy network	–	2×256 MLP	Actor: Linear(d_{obs}, 256)–Tanh– Linear(256, 256)–Tanh– Linear(256, d_{act}).
Value network	–	2×256 MLP	Critic: Linear(d_{obs}, 256)–Tanh– Linear(256, 256)–Tanh– Linear(256, 1). Surface training uses one fleet value; underwater training sums per-agent values.
Per-agent observation dimension	d_{obs}	16/28/40/52; 19	Surface observations for 4/8/12/16 SUVs are 16/28/40/52, respectively. The four-SUV underwater IPPO evaluation uses $d_{\text{obs}} = 19$.
Surface observation	–	Wind at 10 m above sea level	Using ideal surface communication, the fleet forms one shared vector of wind speed and components at 10 m above the sea surface from all SUVs. The common surface policy query also receives the common normalized ground velocity and $[\cos\psi, \sin\psi]$.
Underwater observation	–	Current only	With no inter-vehicle communication, the active IPPO policy observes only its own local current speed and components at the assigned ocean depth, together with a five-frame history, its own normalized propulsion velocity, and $[\cos\psi, \sin\psi]$.
Action dimension	d_{act}	2 per fleet; 2 per SUV underwater	The surface policy emits one continuous east–north joint formation command in $[-1, 1]^2$ and applies it to every SUV. Underwater, each SUV receives its own two-dimensional command. Circular projection enforces the speed bound.
Surface formation constraint	–	Hard shared action	All surface SUVs execute the same action and retain fixed initial relative offsets throughout an episode.
Rollout length	n_{steps}	2048	Number of steps collected per environment before each PPO update.
Mini-batch size	B	256	Mini-batch size used during each gradient update.
Optimization epochs	N_{epochs}	10	Number of passes over each rollout.
Learning rate	α	3×10^{-4}	Adam learning rate used by the actor and critic.
Discount factor	γ	0.99	Reward discount factor.
GAE parameter	λ_{GAE}	0.95	Generalized advantage estimation applied to the team-average reward.
Policy clipping range	ϵ	0.2	Clipping range of the PPO probability ratio.
Value clipping range	ϵ_v	0.2	Clipping range of the value-function update.
Entropy coefficient	c_{ent}	0.01	Entropy regularization coefficient.
Value-loss coefficient	c_v	0.5	Weight of the value-function loss.
Maximum gradient norm	$g_{\max}$	0.5	Global gradient clipping threshold.
Training steps	T	2×10^6	Nominal environment steps per training run.
Training seeds	–	0–9	Ten random seeds for each selected surface or underwater IPPO configuration.
Surface training forecast-error radius	R_{train}	0–800 km	The candidate surface formation-center radial offset is sampled uniformly in $[0, 800]$ km, with a uniformly sampled bearing. This models the assumed upper bound of TC track-forecast error; offshore-only rejection sampling is then applied as part of the safety layer.
Surface validation center offsets	R_{test}	100/200/400/600/800 km	Fixed surface formation-center offsets with random bearings, evaluated without retraining. These values represent increasing track-forecast displacement.
Underwater initialization radius	$R_{0,\text{u}}$	≤200 km	At the end time of each surface segment, the separate subsurface episode initializes four vehicles with radial offsets sampled uniformly in $[0, 200]$ km and uniformly sampled bearings. Surface endpoint positions are not transferred. The 0–800 km forecast-error protocol is surface-only.
Decision interval	Δt	10 min	Environmental observation and action interval.
Initial formation mean radius	R_{form}	100 km	Ring, line, grid, and V layouts are normalized to the same mean radius.
Surface fleet sizes	N	4/8/12/16	Number of SUVs in the final surface configurations.
Underwater fleet size	N	4	Number of SUVs in each underwater comparison case.
Underwater depths below the sea surface	z	0/10/20/30/40/50/60 m	Seven ocean-current observation depths used in the final subsurface matrix.
Underwater methods	–	IPPO; PF; A*; lawn; greedy	IPPO and four traditional planners evaluated on the same episodes and deployments.
Underwater duration	H	20 h	Fixed simulation horizon for the final subsurface comparison; this is not an endurance specification.
Surface ground-speed limits	$V^{\text{s}}_{\max}$	12 or 20 m/s	Two commanded ground-speed envelopes; these are modeled scenarios, not independently measured certifications.
Submerged speed limits	$V^{\text{u}}_{\max}$	1.5 or 2.0 m/s	Two propulsion-speed limits after unit-disk action projection used for underwater IPPO evaluation.
Compute device	–	CPU	All reported IPPO experiments were trained on CPU.

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